Phonetic reduction as a function of referential accessibility: Kinematic evidence from Turkish Sign Language

Word count: 12,558

Abstract Theories of referential accessibility hold that people use reduced linguistic forms for accessible referents, an economy principle standardly studied through the choice among referring expressions. This study asks whether the same principle governs phonetic implementation in Turkish Sign Language (TİD): whether discourse context and continuous referential accessibility predict how nominal referring expressions are articulated, and whether age of acquisition (AoA) modulates these effects. Twenty-eight deaf signers, deaf-of-deaf (DoD) and deaf-of-hearing (DoH), retold ten cartoon narratives; for two high-frequency nominals (CAT, MOUSE), we quantified sign duration and two pose-estimation-derived kinematic measures (hand distance and signing space size) at the sign and utterance levels. All three sign-level measures track discourse status and accessibility: introductions are longer, involve more movement, and occupy more signing space than later mentions, and higher accessibility brings gradient reduction. Duration registers finer accessibility distinctions (including the reintroduction–maintenance region) than the spatial measures, which mainly mark a new-versus-predictable contrast. AoA affected neither the overall level of these sign-internal measures nor their accessibility-driven reduction slopes; DoD signers instead produced shorter event retellings than DoH signers, without kinematic differences, which may suggest that delayed first-language exposure affects referential selection rather than phonetic execution, as predicted by interface accounts of acquisition. The results provide cross-modal evidence for a discourse–phonetics interface: economy of effort shapes not only referring-expression choice but the fine-grained kinematics of signed articulation.

Keywords: Turkish Sign Language; phonetics; discourse; referential accessibility; kinematics

1 Introduction

Effective communication in any language necessitates a balance between clarity and brevity, a principle often justified by cognitive frameworks that favor efficiency and linguistic economy (Zipf 1949; Grice 1975; Sperber & Wilson 1986). In the domain of discourse, this drive for economy is manifested in discourse cohesion strategies, where the selection of linguistic forms is shaped by the assumed cognitive accessibility or activation of a referent in the mind of the addressee (Ariel 1990; Givón 1983). While it is well-established that speakers and signers use more attenuated or reduced forms (e.g., pronouns or null forms) for highly accessible referents and more explicit forms (e.g., full noun phrases) for less accessible ones (Ariel 1990; Gundel et al. 1993; Ferrara et al. 2023), less is known about how these discourse-pragmatic demands affect fine-grained phonetic implementation in sign languages. Recent work on TİD has begun to close this gap for information structure: Gürer et al. (2025) show that narrowly focused signs are lengthened relative to non-focused and broad-focus counterparts, with complementary duration compression in the pre-focal or post-focal domain. Whether comparable

phonetic modulation reflects *referential accessibility* across discourse, rather than focus prominence alone, remains an open question.

The present study investigates how principles of efficiency are realized in Turkish Sign Language (TİD) by examining the relationship between discourse-level factors (discourse context and referential accessibility) and phonetic reduction in lexical referring expressions (REs). Sign language phonetic reduction has previously been examined in terms of spatial lowering or duration shortening in frequent or predictable contexts (Tyrone & Mauk 2012; Hoetjes et al. 2014). However, recent advancements in computer vision, particularly markerless pose estimation tools like MediaPipe (Lugaresi et al. 2019), now allow for the large-scale quantification of kinematic measures such as hand distance and signing space size from standard video data (Börstell 2023).

Whereas previous work on TİD has described which referential forms occur in which discourse contexts [reference removed for anonymous review], the present study asks how those forms are articulated. By applying this computational approach to two selected referential expressions in a story-telling production experiment, we pursue four specific research objectives. Phonetic effort is operationalized throughout as sign duration (from annotated boundaries) together with two MediaPipe-derived kinematic measures: hand distance and signing space size. First (*Discourse-context effects*), we ask whether discrete discourse context (introduction, maintenance, re-introduction) systematically predicts these measures, such that nominal REs in introduction contexts are longer, involve more movement, and occupy more signing space than later mentions. Second (*Accessibility-driven reduction*), we test whether a continuous accessibility score, computed for each RE from distance to the last mention, topicality, and competition among referents (adapted from Toole 1996; see Section 3), predicts continuous phonetic reduction, with higher scores corresponding to less articulatory effort. Third (*Temporal vs. spatial reduction*), we examine whether duration and the two kinematic measures respond to accessibility at different levels of granularity, given that temporal and spatial-articulatory dimensions may be modulated differently by discourse predictability. Fourth (*Age-of-acquisition effects*), we ask whether age of acquisition affects discourse-driven phonetic reduction at the sign level, and whether deaf-of-deaf (DoD) and deaf-of-hearing (DoH) signers differ in overall temporal efficiency at the utterance level, measured as the duration of event retellings.

The remainder of the paper is organized as follows. Section 2 reviews the background on sign language phonetics and discourse reference, referential accessibility and phonetic economy, age-of-acquisition effects on discourse reference, and computer-vision approaches to quantifying sign kinematics. Section 3 describes the participants, narrative elicitation materials, discourse annotation, MediaPipe-based landmark extraction, phonetic measures, and statistical analyses. Section 4 reports the results for discourse-context effects, accessibility-driven reduction, and AoA comparisons at the sign and utterance levels. Section 5 discusses the implications of these findings for a discourse–phonetics interface in the visual-manual modality, Section 6 outlines limitations and future directions, and Section 7 concludes.

2 Background

2.1 Sign language phonetics and discourse reference

Like spoken languages, sign languages have an internal phonological structure: each sign is composed of simultaneously specified parameters (handshape, location, movement,

and orientation) alongside non-manual markers such as facial expression and body posture (Stokoe 1960; Brentari 1998; Dikyuva et al. 2017). The three-dimensional space in front of the signer's body, known as the *signing space* (see Figure 1), serves a dual function: it is the primary articulatory workspace within which signs are physically produced, and a discourse-pragmatic resource through which referents can be indexed at spatial loci and tracked across a narrative.

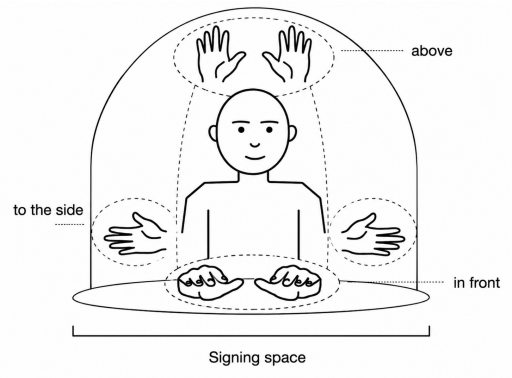


Figure 1: The signing space in sign languages: a quarter-circular area encompassing the head, the space above the head, the torso, the empty space in front of the body, and both sides of the body.

The present study focuses on Turkish Sign Language (TİD), the official sign language of the Deaf community in Türkiye under Law No. 5378 (2005), which has been in use since at least 1889 (Taşçı & Göksel 2018) and has its own phonology, morphology, and syntax described systematically since the early 2000s (Zeshan 2003; Dikyuva et al. 2017). To our knowledge, this is the first study to examine discourse-driven phonetic reduction in TİD using kinematic measures. The examples below illustrate the referential tools available in TİD narratives (see the Abbreviations section for gloss conventions). The linguistic inventory is rich and some of it is modality-specific: it includes *nominals* (full lexical signs such as CAT or MOUSE; see (1)),

(1) Bare nouns in TİD



MOUSE CAT

Pronominal index signs are directional pointing signs that establish a discourse locus in the signing space. Once a referent is assigned a spatial locus, the index sign (IX) can re-introduce it by pointing to that location, as in (2) (Nuhbalaoglu 2018).

(2) Pronominal IX in TİD



IX CA:HIDE.BEHIND.COUCH

'He hides behind the couch.'

Classifier predicates are handshapes that depict the size, shape, or movement of a referent. Whole-entity classifiers (WECLs), for instance, represent an entire referent in space, as in (3) (Sevgi & Gökgöz 2023).

(3) Whole-entity classifier (WECL) in TİD

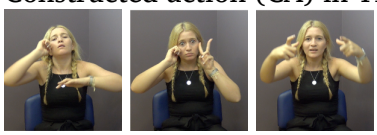


WECL:TWO.ENTITIES.SITUATED

'(The two mice) are situated here.'

Zero anaphora can be realized through constructed action (CA), in which the signer shifts body, gaze, and facial expression to enact a character's actions from a first-person perspective, as in (4) (Arık 2013).

(4) Constructed action (CA) in TİD



CA:SLEEP SEE HAT

'(The mouse) was sleeping, and then he saw the hat.'

Verbal predicates also realize zero anaphora. Plain verbs (uninflected for person or number) and agreement verbs (whose path movement encodes both subject and object agreement by starting and ending at their referential loci) allow arguments to be omitted when the referents are sufficiently accessible. Example (5) illustrates each type.

(5) Verbal predicates in TİD

a. Plain verb



SICK BECOME

'(The cat) became sick.'

b. Agreement verb



$_{1A}$LOOK$_{3B}$

'(The cat) looked at (the mouse).'

In TİD, as in other sign languages, these RE tools differ in explicitness: nominals are the most explicit, followed by pronominal signs, classifiers, and zero anaphora, which is the most reduced and demands the most contextual support from the addressee (Czubek 2017; Frederiksen & Mayberry 2016). This ordering maps directly onto accessibility hierarchies discussed in the following section.

2.2 Referential accessibility and phonetic economy

The relationship between a referent's cognitive status and its linguistic form is central to theories of discourse cohesion. Drawing on the Principle of Least Effort (Zipf 1949) and Grice's (1975) Maxim of Quantity, linguistic behavior is understood as a balance between minimizing articulatory effort and providing sufficient information for the addressee. Accounts of referential accessibility range from givenness-based models (Chafe 1976; Prince 1978) through hierarchical models linking referential form to graded cognitive statuses (Gundel et al. 1993), to topic-continuity approaches in which high continuity licenses reduced forms (Givón 1983). Ariel's (1990) accessibility hierarchy integrates these factors dynamically: accessibility is graded by distance from the last mention, competition with other referents, saliency, and unity, with full noun phrases used for low-accessibility referents and pronouns or zero anaphora for high-accessibility ones. In sign languages, this hierarchy extends to modality-specific forms, with nominal signs at the most explicit end of the continuum (Czubek 2017). Referents in discourse are commonly tracked through a tripartite scheme of introduction, maintenance, and re-introduction (Gullberg 2006; Hickmann et al. 1996). Toole (1996) operationalized this scheme into a graded scale from −2 (least accessible) to 6 (most accessible) based on distance and unity, topicality, and competition.

Phonetic reduction serves as the articulatory manifestation of this linguistic economy. In spoken languages, repeated or predictable references undergo acoustic shortening and articulatory reduction (Bard et al. 2000; Bell et al. 2009). Sign languages also exhibit phonetic reduction through duration shortening and articulatory undershoot (failure to fully reach a sign's canonical location or handshape target, typically under time pressure). Mauk (2003) demonstrated that fast signing produces this undershoot of phonetic targets analogously to fast speech, and Tyrone & Mauk (2012) showed that forehead-located ASL signs are lowered in space as a function of signing rate and adjacent sign context. Stamp et al. (2024b)'s kinematic analysis showed that Israeli Sign Language (ISL) nominal REs for referent introduction involved longer and more spatially expanded articulations, with predictable referents undergoing systematic phonetic reduction across all measures. Within TİD, Gürer et al. (2025) further demonstrate that duration is a primary manual cue for focus prominence in the extended prosodic domain, with focused signs lengthened and neighboring non-focused signs compressed. The current study extends this line of work by examining whether these reduction patterns are sensitive to *continuous* accessibility scores rather than only the tripartite discourse classification or focus marking.

2.3 Age of acquisition and discourse reference

Prior work on TİD has shown that AoA affects multiple linguistic domains, including morphosyntax, verbal fluency, and spatial language (Karadöller et al. 2017; Kayabaşı & Gökgöz 2023; Sevgi & Gökgöz 2023). The vast majority of deaf children (approximately 90%) are born to families with hearing parents who have little or no experience with a sign language, resulting in delayed first language exposure (Mitchell & Karchmer 2004). First systematic input often begins only when these children enter formal deaf education (Mayberry 2007). Previous research has consistently shown that this delay in early access to a systematic sign language negatively affects complex morphosyntactic structures, and that these effects persist into adulthood (Boudreault & Mayberry 2006; Cheng & Mayberry 2021; Sevgi & Gökgöz 2023; Kayabaşı & Gökgöz 2023).

In the domain of discourse reference, Deaf-of-Hearing (DoH) signers and hearing L2 learners of sign languages tend toward *over-explicitness*: they use more overt referential

forms for contextually accessible referents than native signers do (Bel et al. 2015; Frederiksen & Mayberry 2019; Cormier et al. 2013). In TİD specifically, however, a previous study of reference tracking in TİD [reference removed for anonymous review] found that this difference was limited and context-dependent: DoH signers showed modest over-redundancy in maintained and introduced contexts while otherwise approaching native-like competency in simple narratives. The key contrast was that DoD signers were more sensitive to pragmatic economy, leveraging zero anaphora more readily when referents were highly accessible, particularly for maintained references. The Interface Hypothesis (Sorace 2011) provides a theoretical account: grammatical structures requiring the integration of narrow syntax with discourse-pragmatic context are particularly vulnerable to delayed first-language exposure, even when core phonological and morphosyntactic knowledge appears intact. This implies an asymmetry: it is the *selection* of referential form in response to discourse context that is vulnerable, not the phonological and articulatory execution of whichever sign is chosen. Consequently, DoH signers may use slightly more nominals at highly accessible referent contexts, but the *internal* kinematic properties of those nominal signs should still be modulated by discourse context in the same way as for DoD signers. This dissociation also motivates looking beyond individual signs to the utterance. Because nominals are temporally costlier than reduced alternatives (CA, classifiers, or zero anaphora), a higher rate of over-explicit nominal selection, even if modest, can accumulate across a narrative as longer event retellings for DoH than for DoD signers.

2.4 Quantifying sign language phonetics via computer vision

Advances in computer vision provide new opportunities to test theories of linguistic efficiency and phonetic reduction on a larger scale (Börstell 2023). While the quantification of sign language kinematics has relied on marker-based motion capture (Tyrone & Mauk 2010; Puupponen et al. 2015), this approach is often logistically constrained. The recent emergence of markerless pose estimation algorithms, such as OpenPose (Cao et al. 2021) and MediaPipe (Lugaresi et al. 2019), has enabled researchers to extract 2D and 3D landmarks directly from naturalistic video data.

Several recent studies have applied markerless pipelines to continuous signed discourse. For instance, Flaherty et al. (2023) used markerless depth-mapping to document generational reductions in the signing space of Nicaraguan Sign Language users. Within TİD, pose estimation frameworks have recently been adopted to analyze phonetic correlates of various grammatical features, including telicity (Yaşar 2025) and pointing signs (Eroğlu 2025). Separately, Stamp et al. (2024b) used Microsoft Kinect motion tracking to examine phonetic reduction in Israeli Sign Language narratives, demonstrating that maintained nominals undergo measurable reductions in duration and spatial size.

Building on this methodological background, the current study uses MediaPipe's pose estimation to derive objective phonetic measures (hand distance and signing space size) at both the sign level (for two nominal referential expressions) and the utterance level (for an age-of-acquisition comparison). By bridging discourse annotation with markerless kinematic tracking, we document where and how articulatory effort is modulated by discourse context, and whether the timing of first language exposure has an effect on these patterns, addressing the four research objectives set out in Section 1.

3 Methods

3.1 Participants

A total of 29 deaf adult signers were recruited. Participants were divided into two groups based on their age of acquisition (AoA) and the hearing status of their caregivers while they were children. We use the terms deaf-of-deaf (DoD) for deaf children of deaf parents and deaf-of-hearing (DoH) for deaf children of hearing parents. One participant initially recruited as DoD was excluded from the final analysis due to primarily using homesign rather than TİD in childhood despite having deaf caregivers (see Appendix A for diagnostic details), leaving 28 signers (14 males, 14 females) in the analyzed sample. The analyzed DoD group ($N = 14$) consisted of signers who had at least one deaf parent and were exposed to TİD from birth. Their ages ranged from 18 to 35 ($M = 27$, $SD = 5$). Their length of education (LoE) ranged from 12 to 14 years ($M = 12.4$, $SD = 0.9$); 11 of 14 had attended deaf schools across all three levels (primary, middle, and high school). The DoH group ($N = 14$) comprised signers with non-signing parents. Their self-reported AoA ranged from 4 to 17 years. Their ages ranged from 20 to 47 ($M = 35$, $SD = 8$). Their LoE ranged from 12 to 22 years ($M = 14.3$, $SD = 2.8$); 8 of 14 had attended deaf schools across all three levels, while the remainder attended a subset of levels or primary school only. While the two groups differed significantly in mean age ($p < .01$), they were matched for self-reported TİD proficiency and years of active language use ($p > .05$). All retained participants were prelingually deaf, with the exception of one DoH signer. Detailed demographics for the analyzed sample are summarized in Table 1. Ethics approval and consent are reported in the Ethics and consent statement.

3.2 Dataset, stimuli, and procedure

The dataset consists of narrative retellings of ten short video clips from the *Tom & Jerry* animated series. Each clip (duration: 17–28s) depicted discrete events involving the two main protagonists (Tom the cat and Jerry the mouse), with a female cat and a black cat appearing as secondary characters in two of the ten clips. Participants watched each clip as many times as desired before narrating the story. Recording sessions were conducted using an HD video camera focusing on the participant's upper body. In some sessions, a deaf research assistant was present as an addressee, and participants narrated the story to that assistant; in the remaining sessions, no addressee was present, and participants narrated directly to the camera. In both conditions, participants were instructed to narrate naturally, as if to a deaf friend, to avoid register adjustment to a non-native addressee. Statistical analysis (see Appendix B) indicated that interaction style (narrating to camera vs. addressee) did not significantly affect continuous accessibility scores. The discourse annotations of this corpus have previously supported a published analysis of referring-expression distributions across discourse contexts [reference removed for anonymous review]; the kinematic measures and all analyses reported below are new to the present study.

3.3 Discourse context and annotation

Following the discourse context annotation criteria by Gullberg (2006), referents of any syntactic role (e.g., subject, object, or adjunct) had the status introduced if they were brought to the discourse context for the first time in the narration of a particular stimulus. For example, if the signer introduced the referent Tom with a bare nominal CAT

No	Group	Gender	AoA	Age	LoE	Attended Deaf Schools	Home Language
2	DoD	Male	0-3	33	12	PS, MS, HS	TİD, homesign
3	DoD	Female	0-3	27	12	PS, MS, HS	TİD, homesign
13	DoD	Female	0-3	27	12	PS, MS, HS	Homesign
14	DoD	Male	0-3	29	12	PS, MS, HS	TİD, homesign
18	DoD	Male	0-3	28	12	PS, MS	TİD
20	DoD	Male	0-3	32	12	PS, MS, HS	TİD
22	DoD	Male	0-3	35	12	PS, MS, HS	TİD
24	DoD	Female	0-3	23	12	PS, MS, HS	TİD
26	DoD	Female	0-3	30	12	PS, MS, HS	TİD
27	DoD	Female	0-3	18	12	-	TİD
31	DoD	Male	0-3	30	12	HS	TİD
33	DoD	Female	0-3	22	14	PS, MS, HS	TİD
36	DoD	Male	0-3	20	14	PS, MS, HS	TİD
38	DoD	Female	0-3	22	14	PS, MS, HS	TİD
4	DoH	Male	4-7	47	14	PS, MS, HS	TR, homesign
5	DoH	Male	4-7	45	16	PS	TR, homesign
7	DoH	Male	4-7	40	12	PS, MS, HS	TR, homesign
10	DoH	Female	4-7	35	14	PS, MS, HS	Homesign, TR
23	DoH	Male	4-7	36	12	MS, HS	TR, homesign
25	DoH	Female	4-7	31	22	PS, MS, HS	Homesign, TR
37	DoH	Female	4-7	20	14	PS, MS, HS	TR, homesign
1	DoH	Female	8-12	43	12	PS, MS, HS	TR, homesign
6	DoH	Female	8-12	39	16	PS, MS	TR, homesign
12	DoH	Male	8-12	28	16	PS	TR
15	DoH	Female	8-12	24	12	PS, MS, HS	TR
16	DoH	Male	8-12	33	12	PS, MS, HS	TR
17	DoH	Female	8-12	38	12	MS, HS	TR, homesign
32	DoH	Male	13-17	31	16	PS	TR

Note. DoD = deaf-of-deaf (deaf children of deaf parents); DoH = deaf-of-hearing (deaf children of hearing parents); AoA = Age of Acquisition (years); LoE = Length of Education (years); PS = Primary School; MS = Middle School; HS = High School; TR = Turkish. One additional DoD recruit (Participant 34) was excluded prior to analysis (Appendix A) and is not listed here.

Table 1: Participant demographics

in the context of a new stimulus narrative, this referent received the introduced context, regardless of the referent being introduced by the same participant in an earlier narrative.[1] Subject referents were coded for maintenance and re-introduction. In other words, referents that were continued across two or more immediate clauses had the maintained context only if they were the grammatical subjects of the clause. In contrast, subject referents were coded as re-introduced if their mention was discontinued across two or more immediate clauses, or if at least one competing referent was mentioned between their previous and current mention, whether that competing mention occurred in an intervening clause or earlier within the current clause (as with coordinated subjects). Mentions of any RE type, including classifier predicates and constructed action, counted as mentions for the purposes of this criterion. All discourse context and RE type annotations were completed by a single trained annotator.

[1] The accessibility scores, the calculation of which is explained in the following subsection, for these two introduced referents will be different, with the second instance having a slightly larger accessibility score.

3.4 Referring Expression distributions and preliminary analysis

As discussed in Section 2.1, TİD employs a rich array of referential strategies in discourse, ranging from full nominals through pronominal index signs and classifier predicates to constructed action (CA) and verbal constructions. Because the present study aims to isolate the phonetic properties of explicit lexical forms, our kinematic modeling is restricted to the nominal category, focusing on the internal phonetic properties of two nominals: CAT (TİD: *kedi*; corresponding to the protagonist Tom) and MOUSE (TİD: *fare*; corresponding to Jerry). These signs were chosen for three reasons: they represent the two main protagonists across all stimuli clips (see Section 3.2), providing a high token count; they are lexically consistent across signers; and restricting analysis to two items controls for phonological variation in handshape, movement, and location, ensuring that observed kinematic differences reflect pragmatic factors rather than inherent differences in sign form.

The general distribution of RE types across discourse contexts in this corpus has been reported previously [reference removed for anonymous review]. The final dataset for the discourse-level analysis included 4,027 REs: 1,057 (26%) were used for introduction, 1,638 (40%) for maintenance, and 1,332 (33%) for re-introduction.

The categorical RE distribution (Figure 2) followed the expected accessibility pattern: nominals were the dominant strategy for introduction ($M = 94.1\%$), while signers primarily shifted to reduced forms during maintenance, with CA and verbs accounting for approximately three quarters of all constructions ($M_{CA} = 44.3\%$, $M_{PlainVerb} = 19.5\%$, $M_{AgrVerb} = 9.9\%$). For re-introduction, nominals returned as the prevalent tool ($M = 64.1\%$). Crucially, group comparisons revealed that DoH signers used significantly more nominals for maintenance compared to DoD signers ($M_{\text{DoH}} = 8.1\%$ vs. $M_{\text{DoD}} = 4.2\%$).

3.4.1 Assigning accessibility scores

After discourse context and RE type had been annotated in ELAN, the annotation files were exported to a comma-separated value (.csv) file containing participant number, acquisition group, clip number, referent name, discourse context, and RE type. Each RE was then assigned a continuous accessibility score using Toole's (1996) accessibility rating scale (p. 273), originally developed for English across four genres (see Czubek 2017 for its implementation in ASL). We adapted this scale to TİD to move beyond the tripartite discourse labels (introduction, maintenance, re-introduction) to a continuous measure of referential accessibility. Toole's scale quantifies the potential difficulty of retrieving a referent within a given discourse context on a scale of 6 (most accessible) to −2 (least accessible). The score is based on three factors: distance and unity, topicality, and competition. (A fully worked example applying all three factors to three consecutive clauses is provided in Appendix C; here we summarize the algorithm.) Our adaptation differs from the original only in the scoring of the distance and unity component and therefore ranges from 5 (most accessible) to −2 (least accessible). One reason for this is that we focus on subject REs for maintenance and re-introduction, following the annotation criteria in the reference tracking literature (e.g., Frederiksen & Mayberry 2016; 2019; Gullberg 2006). Furthermore, the original scale used the term "episode" to mean a set of propositions that compose a semantic unit in a discourse, similar to a paragraph in a written composition. Since the signers in this study narrated clips that were connected but independent (e.g., they have different plots but share the same protagonists), we took each clip narrative to be one episode and rated the unity factor accordingly. The algorithm for each of the factors was as follows:

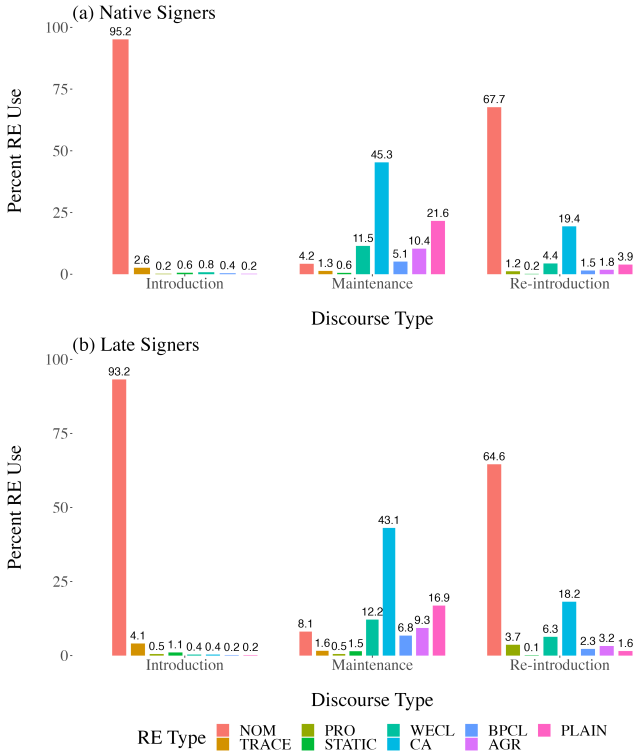


Figure 2: Referring Expression (RE) Type distribution by discourse context.
Note. NOM = Nominal; TRACE = Tracing size and shape specifier; PRO = Pronominal; STATIC = Static size and shape specifier; WECL = Whole-entity classifier; CA = Constructed action; BPCL = Body-part classifier; AGR = Agreement verb; PLAIN = Plain verb.

Distance and Unity: The scoring of distance was determined by the proximity of the last mention and whether referents share the same narrative context or not. This score decreased as the distance in textual context between the current and previous mention increased, reflecting the cognitive load required to recall and track the referent, such as the following:

For a referent R at point t in the discourse (R_t):

a. If the last mention of R is in the clause immediately preceding that of R_t, and they are both in the same narrative context, then the accessibility of R_t is 3.
b. If the last mention of R is in the same narrative context as R_t, but not in the same or preceding clause, then the accessibility of R_t is 2.
c. If the last mention of R is in the narrative context immediately preceding that of R_t, then the accessibility of R_t is 1.
d. Otherwise, the accessibility of R_t is 0.

Criterion (a) presupposes unbroken referential continuity between the two mentions: if a competing referent is mentioned between R's previous and current mention (the configuration that triggers re-introduction coding; Section 3.3), the adjacency required by (a) is interrupted and the distance/unity score is assigned by criterion (b) instead. This convention applies uniformly to all RE types.

Topicality: Topicality was evaluated based on how frequently the referent appeared in the discourse context, specifically within the last four clauses prior to the current mention. More frequent mentions increased the score, enhancing the entity's salience and, hence, its accessibility. This increase was by one point for one or two mentions and two points for more than two mentions, as in the following:

For a referent R at point t in the discourse (R_t):

a. If R has been mentioned more than twice in the last four clauses relative to R_t, then the accessibility of R_t is increased by 2.
b. If R has been mentioned once or twice in the last four clauses relative to R_t, then the accessibility of R_t is increased by 1.
c. Otherwise, the accessibility of R_t does not change.

Competition: Competition scores were related to the presence of competing referents that might confuse or dilute the listener's or reader's ability to recall the referent to be tracked. If other referents were mentioned between the last and the current mention of the focal referent, the score was adjusted downwards. This adjustment was made by one point for one competing entity and two points for multiple competitors, reflecting the increased effort in distinguishing between multiple referents. That is:

For a referent R at point t in the discourse (R_t):

a. If there are two or more competing referents between R_t and the last mention of R, then the accessibility of R_t is decreased by 2.
b. If there is one competing referent between R_t and the last mention of R, then the accessibility of R_t is decreased by 1.
c. Otherwise, the accessibility of R_t does not change.

The three factors were applied in the sequence presented above: the base score was initialized from distance and unity, incremented for topicality, and decremented for competition.

As a validity check on the scoring, mean accessibility by discourse context rose from introduction through re-introduction to maintenance (Figure 3), in line with the tripartite reference-tracking scheme.

These discourse annotations provide the independent variables for the kinematic analyses of CAT and MOUSE developed below.

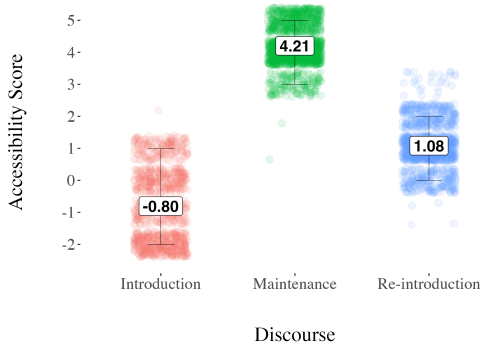


Figure 3: Mean accessibility score by discourse context.

3.5 Landmark extraction and preprocessing

To extract the kinematic features of the annotated nominals, we used the Python (3.12.4) implementation of MediaPipe (0.10.14) (Lugaresi et al. 2019), OpenCV (4.10.0), and NumPy (2.0.0) to detect and track body landmarks. This extraction involves identifying spatiotemporal key points such as wrists, elbows, and shoulders or hands (derived from specific fingers) for both the left and right side of the body. All signers in the videos were right-handed. The 2D videos were processed frame-by-frame; because recordings differed in frame rate, all clips were converted to 50 fps with `ffmpeg` (7.0.1). The input videos were the segmented MOUSE and CAT nouns in the RE analysis, and the annotator-bounded event segments in the utterance-level analysis (Section 3.8). The key points from the signer's body were then tracked, yielding a skeleton representation with numerical coordinates in a three-dimensional space (x, y, z) per frame. The x and y coordinates are normalized image-plane positions (scaled by frame width and height), whereas the z coordinate is not measured by a depth sensor: MediaPipe *estimates* relative depth from monocular RGB input via a learned 3D body model, with the midpoint between the hips as the origin (smaller z = closer to the camera) and a magnitude scaled roughly to x (Lugaresi et al. 2019).

For the landmark extraction and subsequent kinematic analysis, the current study focuses exclusively on the nominal REs CAT and MOUSE, for the item-control reasons detailed in Section 3.4. In total, 23,564 video frames were analyzed (MOUSE: 11,108 frames; CAT: 12,456 frames).

To document the landmark set, Figure 4 shows the 33 MediaPipe pose landmarks used by the framework.

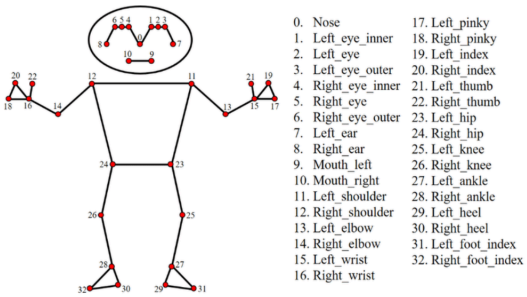


Figure 4: The 33 landmarks used for body pose estimation in the MediaPipe Pose Landmarker (source: Google 2026).

MediaPipe's pose estimation model originally outputs 33 triplets (x, y, z), each of which represents an estimated body landmark, but here we only focus on 10 of them: left thumb, right thumb, left pinky, right pinky, left wrist, right wrist, left elbow, right elbow, left shoulder and right shoulder. Before using these values to calculate phonetic measures, we normalized the extracted triplet data.

During preprocessing, the raw nominal RE data underwent several transformations to ensure reliability and consistency across the dataset. We adopted the following preprocessing steps from previous research (Börstell 2023; Flaherty et al. 2023; Östling et al.; Stamp et al. 2024a; b). First, a median filter (Eq. 6) was applied to the landmark time series: each coordinate value was replaced by the median of the surrounding values within a sliding window. Unlike a moving average, this operation removes brief tracking spikes (outlier jumps caused by the pose estimator or momentary occlusion) while largely pre-

serving the underlying articulatory trajectory, thereby reducing noise without unduly blurring genuine movement:

$$Y_i = \text{median}(X_{i-n}, \ldots, X_{i+n}) \tag{6}$$

where Y_i represents the filtered value, X_i is the original data point, and n is the number of points considered in the filter window.

Because signers differ in body size, raw landmark distances are not directly comparable across participants: a signer with longer arms will produce larger coordinate displacements for the same articulatory effort. To remove this confound, all extracted distance values were normalized to a common bodily scale following Flaherty et al. (2023). For each participant, upper arm length L_i was estimated by averaging the Euclidean distance (Eq. 7) between the shoulder and elbow landmarks on both sides of the body. A participant-specific scaling factor was then computed as $s_i = L_i/\bar{L}$, where $\bar{L}$ is the grand mean upper arm length across all participants. All raw landmark distances for participant i were subsequently divided by s_i. Intuitively, a participant with arms 20% longer than average ($s_i = 1.2$) will have their distances scaled down by that same proportion, while one with shorter arms is scaled up accordingly. After normalization, all distances are expressed in units of the mean upper arm length, placing every signer on an equivalent bodily scale so that observed kinematic differences reflect articulatory behavior rather than anatomical variation.

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \tag{7}$$

3.6 Phonetic measures

Following normalization, we computed the following phonetic measures.

3.6.0.1 Duration.

The interval in seconds (sec) from the first frame (onset) to the last frame (offset) of the RE. A deaf TİD signer, who was not informed about the methodology or research aims, identified these boundaries. Following Stamp et al. (2024b), boundary placement depended on the target's immediate context. When the target nominal followed another sign, the onset was the first frame in which the handshape and orientation of the target became identifiable, so that transitional movement from the preceding sign was excluded; this criterion instantiates the selected finger constraint (Sandler 1999), under which a sign begins once the selected fingers of the target are established. When the target was instead produced directly from a rest position (e.g., hands on the legs), the same onset criterion was applied. The offset was, symmetrically, the last frame before the hands returned to rest or before an identifiable switch in handshape and orientation into the following sign. Duration was then computed as (offset frame − onset frame)/50, given the uniform 50 fps sampling rate.

3.6.0.2 Hand distance.

The total distance the estimated hand position moved (Eq. 7), normalized by duration. Since MediaPipe does not have a specific landmark for "hand," we estimated this hand

position's coordinates by adding the extracted thumb and pinky landmark values and dividing by two for each dimension:

$$(X_{\text{HAND}}, Y_{\text{HAND}}, Z_{\text{HAND}}) = \left(\frac{X_{\text{THUMB}} + X_{\text{PINKY}}}{2}, \frac{Y_{\text{THUMB}} + Y_{\text{PINKY}}}{2}, \frac{Z_{\text{THUMB}} + Z_{\text{PINKY}}}{2} \right). \tag{8}$$

We then computed hand displacement between frames t and $t-1$, summed these values within each RE, and divided by RE duration, yielding a rate of hand movement independent of sign length.

3.6.0.3 Signing space size.

We operationalized signing space as duration-normalized change in the wrist-to-body distance across frames. We first estimated a mid-shoulder reference point by averaging the left and right shoulder coordinates:

$$\begin{aligned} X_{\text{MID_SHOULDER}} &= \frac{X_{\text{L_SHOULDER}} + X_{\text{R_SHOULDER}}}{2}, \\ Y_{\text{MID_SHOULDER}} &= \frac{Y_{\text{L_SHOULDER}} + Y_{\text{R_SHOULDER}}}{2}, \\ Z_{\text{MID_SHOULDER}} &= \frac{Z_{\text{L_SHOULDER}} + Z_{\text{R_SHOULDER}}}{2}. \end{aligned} \tag{9}$$

We then calculated the Euclidean distance between the wrist and the mid-shoulder reference point for each frame, and computed the absolute change across consecutive frames:

$$\Delta D_{\text{SHOULDER_WRIST}} = \left| D_{\text{SHOULDER_WRIST}}(t) - D_{\text{SHOULDER_WRIST}}(t-1) \right|. \tag{10}$$

Finally, we summed these changes within each RE and normalized by the RE duration, so that larger values correspond to greater signing space size, while smaller values correspond to more restricted use of space.

3.7 Statistical analysis

We analyzed the `.csv` output from the landmark extraction pipeline in R. All three phonetic measures showed strongly right-skewed distributions, motivating log transformation prior to modeling. Additionally, data points exceeding ±3 times the interquartile range were treated as outliers and removed, constituting 2% of the nominal RE data (yielding 1,033 observations: MOUSE: 468; CAT: 565). Of the retained 1,033 nominal REs, 404 were introduced, 69 maintained, and 560 re-introduced. We fitted linear mixed-effects models using `lmer` from `lme4` (Bates et al. 2015), with random intercepts for participants and stimulus clips to account for repeated observations within signers and clips. To assess AoA effects at this level, we additionally refitted the accessibility models with signer group (sum-coded) as a main effect and in interaction with accessibility (Section 4.3). The random-effects structure was intercept-only: we did not include by-participant or by-stimulus random slopes for discourse context, accessibility, or RE type. Fixed-effect structure was selected by building upward from simpler to more complex nested models and comparing them with likelihood ratio tests (LRTs). For each phonetic

measure, we began with a single-predictor model (discourse context, or accessibility), then tested whether adding RE type improved fit, and finally whether an interaction between the discourse/accessibility predictor and RE type was warranted; non-significant additions were dropped. On this basis, duration retained Discourse × RE in the discourse analysis and Accessibility + RE in the accessibility analysis; hand distance retained only Discourse (or Accessibility); and signing space size retained Discourse + RE (or Accessibility + RE). The primary retained models can be written as:

```
Measure ~ Discourse + RE + Discourse:RE + (1|Participant) + (1|Stimulus)
```

and

```
Measure ~ Accessibility + RE + (1|Participant) + (1|Stimulus)
```

where `Measure` refers to each of the three log-transformed phonetic outcomes (with measure-specific simplifications as just noted). Categorical predictors in the primary models were sum-coded. We followed up Discourse effects with estimated marginal means and Tukey-adjusted pairwise contrasts using `emmeans` (Lenth & Piaskowski 2026): for the duration model retaining Discourse × RE, testing discourse differences within each RE and RE differences within each discourse context; and for the hand-distance and signing-space models, testing pairwise Discourse contrasts (averaged over RE where applicable). Because accessibility scores and discourse-context labels draw on overlapping discourse properties, we acknowledge a degree of non-independence between the two predictor sets; both are reported to provide complementary perspectives (categorical and gradient), and we address this circularity further in the Limitations. To test whether the accessibility–phonetics relationship is linear or exhibits turning points corresponding to discourse-level boundaries, we additionally fitted polynomial mixed-effects models with quadratic and cubic terms (Appendix D).

3.8 Event analysis

To assess whether AoA affects overall temporal efficiency in narrative production (beyond the RE-level kinematic patterns described above), we conducted a complementary utterance-level analysis using events as the unit of analysis. An event is defined here as a semantically coherent action unit within a narrative clip (e.g., *Tom pulls the carpet back*). Because the stimuli contained many events of varying salience and memorability, event selection followed two criteria to ensure comparability across signers.

First, events were scored for recall completeness: explicit narrations received a score of 1, partial narrations (missing key details) received 0.5, and skipped events received 0. Only events with a mean recall score of ≥ 0.9 across participants were retained. Second, to establish reliable temporal boundaries for duration measurement, two independent annotators (both non-native TİD signers, blind to the experimental design) marked the start and end frames of each event. Events were retained only if the two annotators differed by ten frames or fewer on both the start and end timestamps (at 50 fps, this corresponds to a maximum disagreement of 200 ms). Inter-rater reliability across the retained set was $r = .85$.

Applying these criteria to the 80 possible events across the ten clips yielded 9 unique events suitable for analysis (Events 5, 6, 7 from Clip 2; Events 2 and 4 from Clip 3; Event 9 from Clip 4; Events 4 and 8 from Clip 6; and Event 5 from Clip 10), totaling 233 event retellings across 28 signers. The retained events had recall scores ranging from 92% to

100%, with start/end frame agreement ranging from 97% to 99% across annotator pairs; the 233 retained event retellings had a mean duration of 3.65 s ($SD = 2.68$). For each retained event window, we computed the same three phonetic measures as at the RE level: event duration (s), duration-normalized accumulated hand distance, and duration-normalized signing space size. Landmark extraction, median filtering, and body-scale normalization for these event windows followed the same MediaPipe pipeline described in Sections 3.5–3.6, applied to the event-bounded video segments rather than to the segmented nominal clips. Event duration was log-transformed; as in the RE-level analysis, observations exceeding ±3 times the interquartile range were removed (approximately 3% of the data), leaving 225 event retellings for modeling. Signer group (DoD vs. DoH) was entered as a sum-coded fixed effect, with random intercepts for participant, event, and narrative (stimulus clip). Because the two groups differed in mean chronological age (Section 3.1), a supplementary model added mean-centered age as a covariate to the duration model.

4 Results

4.1 Discourse-context effects

Mixed-effects model results (summarized in Table 2) indicated that discourse context significantly affected RE phonetic realization. Descriptive statistics revealed that mean duration was highest for introduction ($M = 0.54$ s, $SE = 0.012$), compared to re-introduction ($M = 0.38$ s, $SE = 0.007$) and maintenance ($M = 0.34$ s, $SE = 0.018$). Model results confirmed robust introduction-driven lengthening ($\beta = 0.29$, CI: 0.24–0.33, $p < 0.001$); the sum-coded maintenance coefficient was also significant ($\beta = -0.18$, CI: −0.25–−0.11, $p < 0.001$), reflecting shorter maintenance relative to the grand mean rather than a pairwise contrast with re-introduction. These effects are visualized in Figure 5.

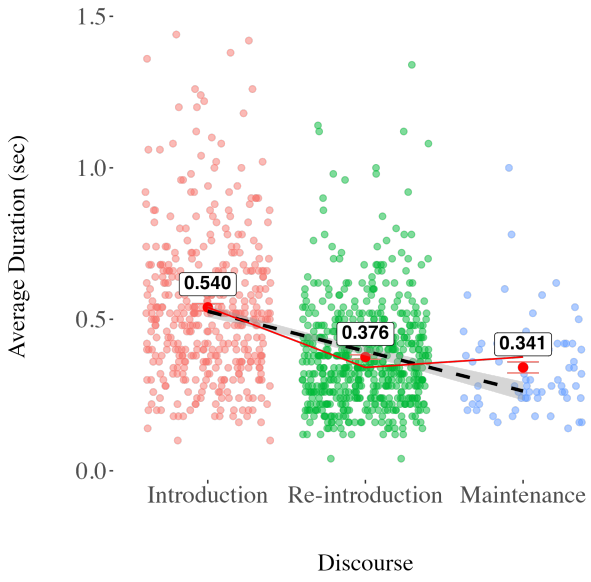


Figure 5: Mean RE duration (s) across discourse contexts. Each point is a single RE; red symbols mark bin means; the dashed line is a linear fit across discourse contexts.

There was also a small main effect of RE type on duration ($\beta = 0.05$, CI: 0.01–0.09, $p = 0.016$), and a significant Introduction × RE [MOUSE] interaction ($\beta = -0.06$, CI: −0.11–

−0.02, $p = 0.008$). The Maintenance × RE interaction was not significant. In pairwise follow-ups, within each RE, introduction was longer than both re-introduction and maintenance for MOUSE (Intro−Reintro: estimate = 0.29, $p < .0001$; Intro−Maint: estimate = 0.34, $p = .0004$) and for CAT (Intro−Reintro: estimate = 0.48, $p < .0001$; Intro−Maint: estimate = 0.58, $p < .0001$; Tukey-adjusted); maintenance and re-introduction did not differ for either sign (both $p > .39$). Thus categorical discourse effects on duration are carried by introduction versus later mentions, not by a reliable maintenance–re-introduction split; introduction lengthening was present for both lexical items, but the discourse contrasts were larger for CAT. Within discourse context, MOUSE and CAT did not differ at introduction ($p = .246$), whereas MOUSE was longer than CAT at re-introduction (estimate = 0.14, $p < .0001$), with only a marginal difference at maintenance ($p = .071$). We return to possible lexical sources of this asymmetry in the Discussion.

For movement-based measures, introduction significantly increased hand distance ($\beta = 0.09$, CI: 0.04–0.14, $p < 0.001$; Figure 6). Hand distance was greatest for introduction ($M = 0.98$, $SE = 0.025$), followed by re-introduction ($M = 0.88$, $SE = 0.017$) and maintenance ($M = 0.81$, $SE = 0.044$). The sum-coded maintenance coefficient was not significant ($\beta = -0.07$, CI: −0.15–0.01, $p = 0.080$), but Tukey-adjusted pairwise contrasts confirmed that introduction exceeded both maintenance and re-introduction, with no Maint–Reintro difference. Similarly, introduced nominal REs increased signing space size ($\beta = 0.06$, CI: 0.01–0.10, $p = 0.009$; Introduction $M = 0.24, SE = 0.005$; Figure 7); the sum-coded maintenance coefficient was not significant ($\beta = -0.03$, CI: −0.10–0.03, $p = 0.332$; Maintenance $M = 0.22, SE = 0.010$), and pairwise tests separated introduction from re-introduction but not from maintenance. The sign MOUSE was associated with a significant reduction in signing space size relative to CAT ($\beta = -0.03$, CI: −0.06–−0.00, $p = 0.023$), a lexical main effect attributable to phonological differences between the two signs (see Discussion) and subject to the MediaPipe depth-measurement caveats noted in the Limitations. Converging descriptively with this spatial pattern, Figure 8 plots the mean articulatory location of the target nominals in each context: maintained nominals cluster proximal to the signer, whereas introduced and re-introduced nominals occupy more distal and varied positions.

Predictors	Duration (log) Estimate [CI]	Hand Distance (log) Estimate [CI]	Signing Space Size (log) Estimate [CI]
(Intercept)	-0.80 [-0.85, -0.74]	-0.05 [-0.09, -0.01]	-1.45 [-1.48, -1.41]
Intro	0.29 [0.24, 0.33]***	0.09 [0.04, 0.14]***	0.06 [0.01, 0.10]**
Maint	-0.18 [-0.25, -0.11]***	-0.07 [-0.15, 0.01]	-0.03 [-0.10, 0.03]
RE [MOUSE]	0.05 [0.01, 0.09]*	N/A	-0.03 [-0.06, 0.00]*
Intro × RE [MOUSE]	-0.06 [-0.11, -0.02]**	N/A	N/A
Maint × RE [MOUSE]	0.04 [-0.03, 0.11]	N/A	N/A
σ^2	0.17	0.22	0.16
τ_{00} Participant	0.04	0.05	0.06
τ_{00} Stimuli	0.01	0.01	0.00
ICC	0.23	0.23	0.26
Marginal R^2 / Conditional R^2	0.166 / 0.358	0.012 / 0.242	0.012 / 0.266

Note. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Discourse and RE were sum-coded; coefficients are deviations from the grand mean, not pairwise contrasts (Tukey-adjusted pairwise tests are reported in the text). N/A = predictor not retained by likelihood-ratio model comparison. The non-significant Maint × RE[MOUSE] duration interaction ($p = .282$) is included for completeness.

Table 2: Mixed-effects model results for RE-level analysis by Discourse context and RE type.

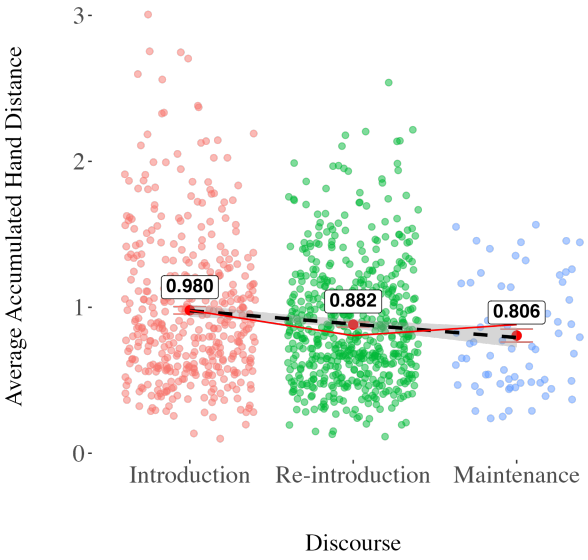


Figure 6: Mean duration-normalized hand distance across discourse contexts. Each point is a single RE; red symbols mark bin means; the dashed line is a linear fit across discourse contexts.

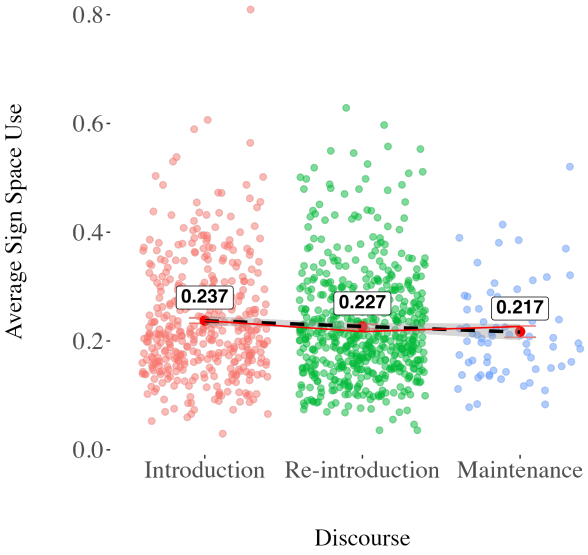
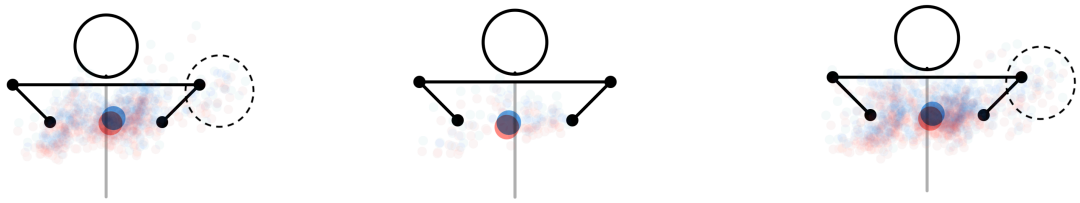


Figure 7: Mean duration-normalized signing space size across discourse contexts. Each point is a single RE; red symbols mark bin means; the dashed line is a linear fit across discourse contexts.

Taken together, the discourse-context models revealed a consistent pattern across all three measures: introduction yielded greater phonetic investment than later mentions, with no reliable maintenance–re-introduction split; finer temporal differentiation between these contexts is taken up in the continuous accessibility analyses below. Maintained nominals were far fewer ($N = 69$) than introduced ($N = 404$) or re-introduced ($N = 560$) REs, so maintenance contrasts should be interpreted with caution.



(a) Introduced (b) Maintained (c) Re-introduced

Figure 8: Average right (red) and left (blue) hand positions for introduced, maintained, and re-introduced nominals. The larger points represent the grand mean for each context. The dashed circles indicate the peripherally signed nominals in the introduced and re-introduced contexts.

4.2 Accessibility-driven reduction

Accessibility showed consistent reduction effects across phonetic measures (summarized in Table 3). For RE duration, higher accessibility scores predicted shorter durations ($\beta = -0.07$, 95% CI: −0.09–−0.05, $p < 0.001$; Figure 9). Accessibility also predicted reduced hand distance ($\beta = -0.03$, 95% CI: −0.06–−0.01, $p = 0.004$; Figure 10) and reduced signing space size ($\beta = -0.03$, 95% CI: −0.05–−0.01, $p = 0.005$; Figure 11).

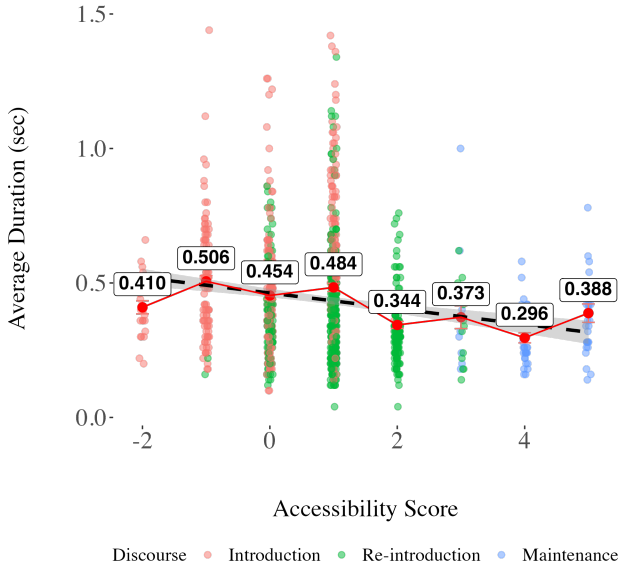


Figure 9: Mean RE duration (s) by accessibility score. Each point is a single RE; red symbols mark bin means; the dashed line is a linear fit across accessibility bins.

To formally characterize the shape of these relationships, we additionally fitted polynomial mixed-effects models. A significant cubic polynomial term for accessibility was found for duration ($\beta = 0.21$, CI: 0.06–0.36, $p = .005$), indicating two turning points along the continuous scale. In contrast, only the quadratic term was significant for hand distance ($\beta = 0.17$, CI: 0.01–0.32, $p = .037$) and for signing space size ($\beta = 0.17$, CI: 0.03–0.30, $p = .014$), with the cubic term non-significant in both cases. This pattern

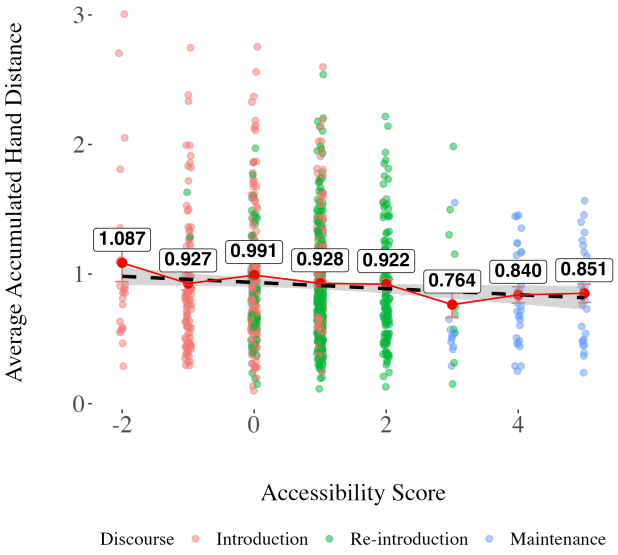


Figure 10: Mean duration-normalized hand distance by accessibility score. Each point is a single RE; red symbols mark bin means; the dashed line is a linear fit across accessibility bins.

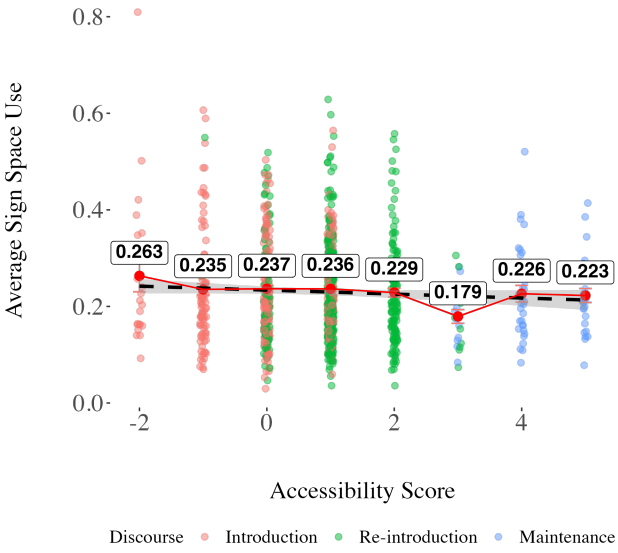


Figure 11: Mean duration-normalized signing space size by accessibility score. Each point is a single RE; red symbols mark bin means; the dashed line is a linear fit across accessibility bins.

was consistent with the descriptive observation that duration dropped by approximately 30% from accessibility score 1→2, a transition that falls within the region of the scale occupied mainly by re-introduced referents (per-context mean accessibility: introduction ≈ -0.8, re-introduction ≈ 1.1, maintenance ≈ 4.2; Figure 3), and by a further 20% from score 3→4, in the transition from the re-introduction toward the maintenance region (Figure 9). This is a finer grain than the categorical Discourse pairwise tests detected. Kinematic measures, by contrast, showed a single notable decline (approximately 17% for hand distance and 22% for signing space size) around the score 2→3 boundary, af-

Predictors	Duration (log) Estimate [CI]	Hand Distance (log) Estimate [CI]	Signing Space Size (log) Estimate [CI]
(Intercept)	-0.84 [-0.88, -0.79]	-0.04 [-0.08, 0.01]	-1.45 [-1.49, -1.41]
Accessibility	-0.07 [-0.09, -0.05]***	-0.03 [-0.06, -0.01]**	-0.03 [-0.05, -0.01]**
RE [MOUSE]	0.03 [0.00, 0.06]*	N/A	-0.02 [-0.05, 0.01]
σ^2	0.20	0.22	0.16
τ_{00} Participant	0.04	0.05	0.06
τ_{00} Stimuli	0.01	0.01	0.00
ICC	0.18	0.23	0.26
Marginal R^2 / Conditional R^2	0.038 / 0.214	0.006 / 0.239	0.012 / 0.266

Note. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The hand distance model did not include RE (N/A); duration and signing space size retained Accessibility + RE.

Table 3: Mixed-effects model results for RE-level analysis by Accessibility Score and RE type.

ter which they stabilized (Figures 10–11). A granular analysis also revealed non-linear trends at the extreme ends of the accessibility scale. Referents with the lowest accessibility ($Score = -2$) exhibited shorter durations than predicted, particularly in sequences of multiple new referents, where the final list item was consistently produced with reduced duration. Conversely, an unexpected increase in duration was observed at the highest accessibility level ($Score = 5$). These two deviations are discussed further in Section 5.

4.3 Age of acquisition and phonetic efficiency

To examine whether Age of Acquisition shaped the phonetic reduction patterns reported above, we refitted the RE-level accessibility models with signer group (DoD vs. DoH) as a main effect and in interaction with accessibility; the interaction is the critical test, since equal reduction requires equal accessibility *slopes*, not merely equal intercepts. Neither the Group main effects (all $p > .22$) nor the Group × Accessibility interactions (duration: slope difference $= -0.03$, $p = .149$; hand distance: < 0.01, $p = .997$; signing space size: -0.01, $p = .606$) were significant: once a nominal was produced, its articulatory properties were modulated by referential accessibility in the same way regardless of AoA. With 14 signers per group, however, these null results warrant the usual caution about statistical power.

A complementary utterance-level analysis (see Section 3.8 for the event selection procedure) assessed whether DoD and DoH signers differed in overall temporal efficiency during narrative production. Table 4 summarizes the mixed-effects model results. Duration was the only measure that differed significantly between groups: DoD signers produced event retellings approximately 0.91 seconds shorter on average than DoH signers (DoD: $M = 3.21$ s, $SE = 0.22$; DoH: $M = 4.12$ s, $SE = 0.27$; $\beta = -0.14$, CI: -0.26–-0.02, $p = 0.020$). As Group was sum-coded, β is the deviation from the grand mean, implying a DoD–DoH difference of -0.27 log units ($\approx 24\%$). By contrast, duration-normalized hand distance (DoD: $M = 1.08$, $SE = 0.03$; DoH: $M = 1.02$, $SE = 0.03$; $p = 0.723$) and signing space size (DoD: $M = 0.28$, $SE = 0.010$; DoH: $M = 0.27$, $SE = 0.009$; $p = 0.885$) did not differ between groups.

Because the DoH group was on average older (Section 3.1), the group contrast is partially confounded with chronological age. Adding mean-centered age to the duration model left the Group estimate unchanged (-0.27 log units; $p = .050$ with the collinearity-

inflated standard error), while age itself predicted nothing ($\beta \approx 0.00$, $p = .99$; age alone: $p = .28$). The group difference is thus not attributable to age, though a design of this size cannot fully deconfound age (or length of education) from AoA.

Predictors	Duration (log) Estimate [CI]	Hand Distance (log) Estimate [CI]	Signing Space Size (log) Estimate [CI]
(Intercept)	1.14 [1.01, 1.26]	0.03 [-0.07, 0.13]	-1.28 [-1.40, -1.16]
Group [DoD]	-0.14 [-0.26, -0.02]*	0.02 [-0.07, 0.11]	-0.01 [-0.12, 0.10]
σ^2	0.20	0.06	0.07
τ_{00} Participant	0.07	0.05	0.08
τ_{00} Event	0.08	0.00	0.02
τ_{00} Narrative	0.19	0.01	0.01
ICC	0.63	0.49	0.64
Marginal R^2 / Conditional R^2	0.035 / 0.642	0.002 / 0.492	0.000 / 0.636
Observations	225	225	225

Note. * $p < 0.05$. Group was sum-coded (deviation from the grand mean; the DoD–DoH difference equals twice the coefficient). Random intercepts for Participant, Event, and Narrative (stimulus clip).

Table 4: Mixed-effects model results for utterance-level analysis by signer group.

These group comparisons are visualized in Figures 12 (duration), 13 (hand distance), and 14 (signing space size).

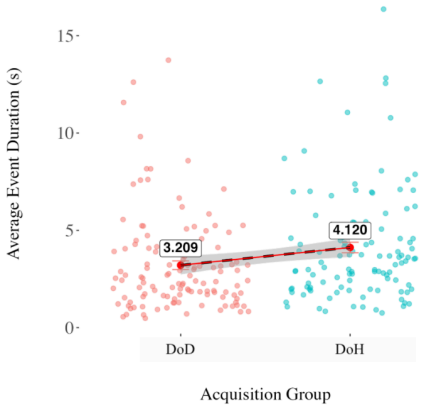


Figure 12: Average event duration (s) by signer group (DoD vs. DoH). Each point is a single event retelling; red symbols mark group means; the dashed line connects the means.

The dissociation between duration and kinematic measures at the utterance level is analogous, though not identical, to the RE-level pattern: temporal measures again prove more sensitive than spatial ones, here to AoA rather than to referential accessibility. Taken together, phonetic reduction in TİD appeared to operate at two levels in these data: a sign-level mechanism driven by referential accessibility that did not differ by acquisition history, and an utterance-level temporal efficiency that was shaped by the depth of early TİD exposure (see Section 5.2).

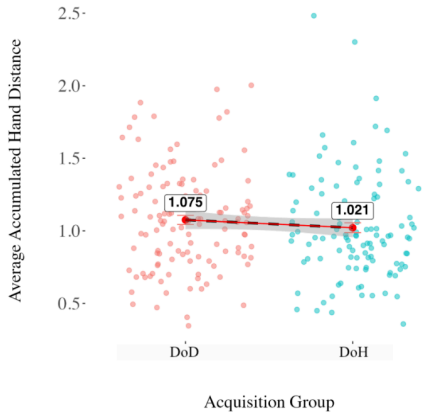


Figure 13: Average duration-normalized hand distance by signer group (DoD vs. DoH). Each point is a single event retelling; red symbols mark group means; the dashed line connects the means.

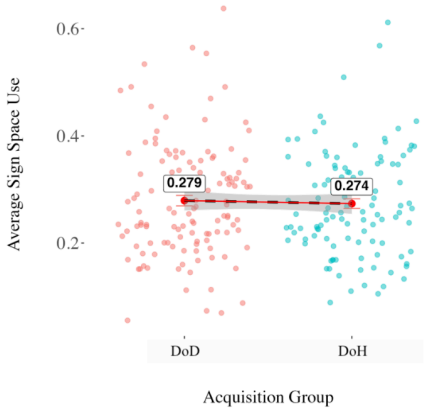


Figure 14: Average duration-normalized signing space size by signer group (DoD vs. DoH). Each point is a single event retelling; red symbols mark group means; the dashed line connects the means.

5 Discussion

The results support a discourse–phonetics interface in which phonetic effort is modulated by communicative demands and information structure. When a referent is newly introduced into the discourse, signers produce longer and kinematically "larger" REs (greater hand distance and signing space size), consistent with the idea that producers invest more articulatory and temporal resources to establish a new discourse entity. Later mentions, however, are shorter and kinematically reduced relative to introductions, though pairwise tests did not reliably separate maintenance from re-introduction in the categorical Discourse models. These patterns indicate that signers dynamically adjust the articulatory prominence of their signs based on discourse status, paralleling given/new shortening in spoken languages (Fowler & Housum 1987; Pluymaekers et al. 2005) and extending previous observations of phonetic reduction in sign languages to TİD (Flaherty et al. 2023; Hoetjes et al. 2014; Tyrone & Mauk 2010; 2012; Mauk & Tyrone 2012). They also complement Gürer et al. (2025), who show that TİD uses duration as a fo-

cus cue (boosting on the focused sign and compression in the extended domain). The present findings indicate that duration likewise tracks referential accessibility in narrative discourse, even outside an experimentally elicited focus paradigm. These results align with Haspelmath (2008)'s argument that formal asymmetries are best explained by frequency-induced predictability and the economy principle, which favors shorter forms for more predictable elements. Crucially, hand distance and signing space size patterned mainly as an introduction-versus-later contrast, consistent with Stamp et al. (2024b) for ISL, who also found that kinematic measures primarily distinguish new from predictable information rather than tracking finer-grained discourse distinctions.

A further question is *why* accessible referents are reduced. Spoken-language research distinguishes two families of explanation (Gahl et al. 2012; Bell et al. 2009; Bard et al. 2000). On *intelligibility-based* (listener-oriented) accounts, producers actively manage the addressee's recognition needs: when discourse context makes a form highly predictable, the signer can afford a less fully articulated token because the addressee can recover the intended referent from top-down cues. On *production-based* (speaker-oriented) accounts, reduction falls out of lexical access and motor execution: highly activated or frequent forms are retrieved and articulated more rapidly, yielding temporal compression and articulatory undershoot without intentional audience design. Recent work on conversational speech further underscores that not all reductions are of one kind: some are relatively stable, lexicon-linked patterns, whereas others are more opportunistic and tightly tied to speaking rate and speaker-specific articulatory habits (Bodur et al. 2025). Sign language research has extensively documented rate-, context-, and style-conditioned reduction (Mauk 2003; Tyrone & Mauk 2012; Stamp et al. 2024b), but has rarely framed discourse-driven kinematic reduction in terms of this listener- versus speaker-oriented contrast. The present TİD results are compatible with either approach: high accessibility could license reduced forms because the addressee can recover the referent, or simply accelerate the retrieval of an already activated nominal. The current design does not allow us to disentangle the two interpretations but the visual-manual modality makes the contrast especially interesting for future work, because the articulators are continuously visible to the addressee, concurrent multi-articulator strategies are available, and communicative pressure may therefore interact with motor economy differently than in speech. We will return, in Limitations, to a proposal for how these alternatives can be examined in future work.

Regardless of its cause, i.e., intelligibility- or production-based, the interaction of discourse context and phonetic reduction provides kinematic evidence from the visual-manual modality for a usage-based, economy account of linguistic production. These patterns align with the "Uniform Information Density" (UID) hypothesis (Levy 2008), which posits that producers maintain a constant rate of information by reducing duration and effort for more probable elements. More broadly, the results extend the classical efficiency frameworks reviewed in Section 1 (Zipf, Grice, Sperber & Wilson) to the physical articulation of signs, corroborating gradient models of referential accessibility (Ariel 1990; Gundel et al. 1993; Tily & Piantadosi 2009).

These discourse-driven effects were not, however, uniform across lexical items. The significant Introduction × RE interaction for duration, together with the pairwise follow-ups, suggests that sign-internal phonological constraints modulate the *magnitude* of discourse-driven lengthening rather than whether lengthening occurs at all. Introduction was longer than re-introduction and maintenance for both MOUSE and CAT; what differed was the size of those contrasts, which were larger for CAT. Critically, the two signs did not differ in duration within introduction itself, so the interaction should not be read as

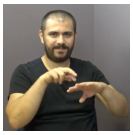
MOUSE failing to lengthen when newly mentioned. The two signs differ in several phonologically relevant ways. In CAT, the primary movement involves a forearm rotation that changes the orientation of the dominant hand; this rotational movement is relatively complex and provides more temporal and spatial scope for discourse-driven lengthening. In MOUSE, the canonical movement is a fast, repetitive trilling of the wrist or a slight orientation-changing movement, both of which are shorter and more constrained in their range. Additionally, in MOUSE (but not in CAT), signers may optionally drop the nondominant hand, which serves as the place of articulation for both signs. Together, these phonological differences plausibly compress the available range of temporal expansion for MOUSE relative to CAT, yielding smaller introduction-versus-later contrasts even though both signs still lengthen at first mention. These observations highlight that phonetic implementation (reduction or increase) operates within the bounds set by a sign's phonological specification.

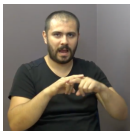
Importantly, the phonetic effects of discourse context were substantially larger than those of RE type. Introduction lengthened duration by roughly 34% for MOUSE and 62% for CAT relative to re-introduction (a difference of about 160 ms in the raw means, 0.54 s vs. 0.38 s), whereas the two signs differed from each other by only about 11% overall (Table 2). This asymmetry is also reflected in the distributional data: a regression of accessibility scores on discourse context labels explained approximately 88% of the variance ($R^2 = 0.877$), compared to only 52% when using RE type as the sole predictor ($R^2 = 0.521$). In categorical Discourse pairwise tests, our duration data likewise did not separate maintenance from re-introduction. What the continuous accessibility and polynomial analyses add is a finer temporal grain: duration continues to decline across accessibility regions that correspond to re-introduction and maintenance, even though those two categorical labels are not pairwise-distinct once Tukey-adjusted. This graded temporal pattern, a functional modularity in how articulatory dimensions are modulated, supports treating duration as the more sensitive phonetic channel, registering finer accessibility distinctions at lower articulatory cost than spatial shifts (Stamp et al. 2024b).

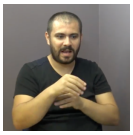
5.1 Deviations at the extremes of the accessibility scale

While the overall accessibility–reduction relationship was robust, two deviations at the extremes of the scale, noted in Section 4.2, warrant further discussion. The divergence at accessibility score −2 (the least accessible referents) may reflect *list prosody* rather than accessibility alone (Selting 2007; Erickson 1992). In spoken listing routines, speakers use duration and other prosodic cues to organize sequential structure: later items, especially those that close the list, are often shortened relative to earlier ones, independently of their semantic newness. A score of −2 in our data arises precisely in such a configuration: a referent that is the third introduction in a sequence of introductions. For example, in the STAIRS–CARPET–MOUSE sequence (see Example 11), MOUSE consistently showed shorter duration despite being semantically new and discourse-inaccessible. We interpret this tentatively as a list-completion effect: rather than expanding articulation in response to low accessibility, the signer shortens the final list item to mark structural closure of the sequence. This would be a prosodic, not pragmatic, reduction. Whether TİD systematically marks list structure through prosody, as spoken languages do, remains an open question for future research.

(11)


STAIRS


CARPET


MOUSE

Discourse Context:	Introduction	Introduction	Introduction
RE Type:	Nominal	Nominal	Nominal
Distance and Unity Score:	0	0	0
Topicality Score:	0	0	0
Competition Score:	0	-1	-2
Total Accessibility Score:	0	-1	-2

Conversely, the unexpected increase in duration at accessibility score 5 (highly accessible referents) requires a different account. Score-5 observations are sparsely populated in the kinematic sample (only 27 CAT/MOUSE REs), so the uptick is partly attributable to sampling instability at the extreme of the scale. More substantively, a full nominal is already marked for such highly accessible referents, which are usually tracked with CA or verbs. When a nominal does appear at this accessibility level, qualitative inspection of the data suggests that lengthening often reflects communicative hesitation, self-correction, or clarifying restatement rather than accessibility-driven expansion, the same unmodeled constraints that produce occasional "long maintenance" outliers elsewhere in the dataset (see Section 6). Critically, over-explicitness in RE *choice* (DoH signers used nominals for score-5 referents three times as often as DoD signers: 12% vs. 4%) cannot explain why those nominals are longer once selected: the present analysis measures only the articulatory properties of CAT and MOUSE, with no comparison to the reduced alternatives that were not produced, and group was not a significant predictor of RE-level duration. Over-explicit nominal selection is instead expected to inflate *utterance*-level narration length, an effect reported in Section 4.3.

5.2 Age of acquisition: selection versus execution

Taken together, the AoA findings reveal a dissociation between two analytic levels. At the *sign level*, we measure the internal phonetic properties of individual nominals once they have been produced (duration, hand distance, signing space size). At the *utterance level*, we measure the overall duration of event retellings, which can reflect discourse packaging and referential-form choice rather than how any single sign is articulated. Under the Interface Hypothesis (Sorace 2011), delayed exposure selectively impairs the discourse-pragmatic decision of *which* form to use, not the articulatory execution of a form once selected. Consistent with that asymmetry, discourse-driven reduction at the sign level shows no DoD–DoH difference: both groups modulate duration, hand distance, and signing space size in response to referential accessibility in the same way (Group × Accessibility interactions all non-significant; Section 4.3). This pattern differs sharply from what Flaherty et al. (2023) document for Nicaraguan Sign Language (NSL), an emerging system in which signing-space size itself continues to shrink across successive acquisition cohorts. In that case, age/cohort of exposure reshapes the spatial-articulatory grammar of the language; in the present TİD data, by contrast, hand distance and signing space size show no DoD–DoH difference at either the RE or the utterance level. What AoA affects is discourse-level packaging: DoD signers produce event retellings approximately

0.91 seconds shorter than DoH signers, without a corresponding difference in hand distance or signing space size. This temporal advantage plausibly reflects greater discourse-structural efficiency, in particular less nominal redundancy in highly accessible contexts, rather than a difference in raw articulatory speed or motoric skill (Börstell et al. 2024). We did not directly measure transitions between the signs in these event windows, which include predicates, classifiers, and constructed action alongside referring expressions, so denser narrative packaging remains a compatible but untested contributor. Sign-level phonetic reduction and utterance-level temporal efficiency thus reflect different aspects of linguistic competence: the former is tied to the phonetic grammar of TİD (already conventionalized, unlike emerging NSL), the latter to the pragmatic and discourse-structural norms that appear more sensitive to the timing of early language acquisition. The over-explicitness observed at the categorical level (DoH signers using nominals for highly accessible referents where DoD signers shift to reduced forms; [reference removed for anonymous review]) is consistent with this account: DoH signers may retain accurate phonological knowledge of how to produce a nominal sign but require greater cognitive resources to judge *when* a nominal is discourse-appropriate, leading to over-use of explicit forms for highly accessible referents. Critically, this vulnerability operates at the level of expression *selection* rather than phonetic *execution*: once a nominal is chosen, its articulatory reduction is governed by the same accessibility-driven mechanism operating in DoD signers.

5.3 Articulatory and prosodic controls

Three potential confounds were identified and addressed. First, introduced signs encompassed both subject and object/adjunct referents while maintenance and re-introduction were restricted to subjects, creating potential positional asymmetries across discourse contexts. This asymmetry cannot be fully eliminated under the present coding scheme, but its impact is partly constrained by the fact that target nominals overwhelmingly precede the verb in TİD narrative discourse, so they are not concentrated in phrase-final position. Second, co-articulatory onset shifts (e.g., spatial assimilation of the target sign to the locus of a preceding inflected sign; see Figure 15) risk conflating transitional movement with the target sign's own articulation. This confound was addressed by identifying sign boundaries using the selected finger constraint (Sandler 1999), a phonological principle that defines a sign's onset as the moment the hand assumes its target handshape rather than the start of the transitional movement from the preceding sign; co-articulatory shifts are thereby excluded from the measured window to a certain degree as co-articulation is still possible due to location assimilation. Third, phrase-final lengthening (Edwards et al. 1991; Wilbur et al. 2012; Tyrone & Mauk 2010) was likewise mitigated by preverbal placement of the target nominals.



Figure 15: Co-articulation and position effect of the preceding sign on the target sign in re-introduction context.

5.4 Modality-specific reduction and kinematic markers

The finding that both duration and signing space size are sensitive to accessibility suggests that signers utilize multiple cues to manage communicative efficiency. While duration is a cross-linguistic cue, the use of space as a phonetic cue is unique to the visual-manual modality. The reduction in signing space size for highly accessible referents represents an "articulatory undershoot" (Lindblom 1990) in the spatial domain, where the target location or movement range is contracted. Beyond the sequential reduction patterns documented here, sign languages possess an additional economy strategy unavailable to spoken languages: concurrent forms. Because visual-gestural languages can leverage parallel articulators (body, face, and hands), signers can simultaneously track one referent through CA while representing another via a classifier handshape (Cormier et al. 2013; Dudis 2004; Janzen 2004). Although we did not analyze concurrent forms in the present study (see Limitations), their existence underscores that the economy pressures documented here operate alongside modality-specific affordances that further amplify communicative efficiency.

Methodologically, the analyses demonstrate that markerless pose estimation (MediaPipe) can be integrated with mixed-effects modeling to operationalize phonetic effort in sign languages using ordinary 2D video. While such pipelines do not replace dedicated motion capture, they provide a relatively low-cost and scalable approach for quantifying temporal and spatial properties of signing, enabling tighter links between discourse-pragmatic constructs (e.g., accessibility) and fine-grained kinematic outcomes.

6 Limitations and future work

Several limitations qualify the present findings. First, the kinematic measures come from markerless pose estimation on 2D video: depth is estimated rather than measured, and tracking can be disrupted by occlusion. This matters especially for MOUSE, whose trilling movement is more depth-dominant than the largely horizontal movement of CAT, so MediaPipe measurement error may be systematically greater for MOUSE and may contribute to its smaller signing-space values. Optical motion capture would be the appropriate validation. Second, the analysis is restricted to two right-handed nominals and to sequential subject REs; broader lexical and structural sampling is needed, and concurrent multi-articulator forms (Dudis 2004; Janzen 2004) remain outside the present kinematic scope. Third, unmodeled factors (animacy (Ferrara et al. 2023), topic-position prosody, hesitation/self-correction, and non-manuals) may affect duration independently of accessibility (including the score-5 lengthening discussed in Section 5).

The design also cannot decide whether accessibility-driven reduction is listener-oriented, speaker-oriented, or both (Section 5). A decisive follow-up would hold discourse accessibility constant while manipulating addressee knowledge (shared vs. unshared cartoon context) or clear-versus-casual signing instructions. As can be seen in Appendix B, we examined interaction style only for accessibility scores, not for kinematic slopes; the session-level record of interaction style was not retained in the analysis dataset, and too few signers narrated to an addressee for a well-powered test in any case.

Finally, the two primary predictors are not independent: Toole's accessibility score is built from distance, topicality, and competition (the same properties that partly define introduction, maintenance, and re-introduction), so the predictors share substantial variance. We report both to show that reduction tracks a continuous scale rather than only a categorical switch, but the unique contribution of accessibility above discourse context

should be interpreted cautiously. Cleaner dissociation would require independently estimated predictability (e.g., corpus surprisal; Levy 2008) or experiments that orthogonally manipulate prior mentions and competitors.

7 Conclusion

To our knowledge, this study provides the first quantitative evidence for gradient phonetic reduction in Turkish Sign Language (TİD) based on discourse context and referential accessibility. Using a scalable, markerless computer vision pipeline together with annotated sign boundaries, we demonstrated that signers systematically modulate their articulatory effort, producing longer durations, greater hand distance, and larger signing space size for newly introduced referents, while reducing these measures for highly accessible later mentions. Relating these articulatory outcomes to a continuous accessibility scale revealed a functional modularity: while spatial measures mainly distinguish new from predictable information, duration is modulated more finely along the accessibility continuum, including transitions that categorical Discourse pairwise tests did not resolve.

Together, these findings are consistent with usage-based frameworks of linguistic efficiency (Zipf 1949; Sperber & Wilson 1986), indicating that the drive for cognitive economy shapes not only the syntactic and lexical choice of referring expressions, but also their physical articulation in the visual-manual modality. By integrating markerless pose estimation with discourse-pragmatic frameworks, this approach opens new pathways for objectively investigating the discourse-phonetics interface and articulatory variation across diverse sign languages.

Abbreviations

Glosses are given in SMALL CAPS. Gloss abbreviations: CA = constructed action (role shift); IX = index (pointing) sign; WECL = whole-entity classifier; BPCL = body-part classifier; SASS = size-and-shape specifier; subscripts $_1/_3$ indicate first/third person spatial loci. Other abbreviations: AoA = age of acquisition; DoD = deaf-of-deaf; DoH = deaf-of-hearing; LoE = length of education; RE = referring expression; TİD = Turkish Sign Language (*Türk İşaret Dili*); UID = Uniform Information Density.

Data availability

The discourse annotations with accessibility scores, the sign-level and event-level kinematic measures, the R analysis scripts, and the Python (MediaPipe) extraction notebooks supporting this study are available in an anonymized repository at https://anonymous.4open.science/r/Glossa. Participants consented to the publication of still images from their recordings, and such stills appear in the figures of this article. The full video recordings are not deposited in the repository, as sign language video is inherently identifying and participants consented to its research use rather than to open redistribution; the derived measures in the repository fully support the reported analyses. The elicitation clips are copyrighted material and are described in Section 3.2. The repository will be made public with a citable DOI upon acceptance.

Ethics and consent

The study protocol was approved by the institutional review board for research with human subjects at [institution name removed for anonymous review] (application no. removed for anonymous review), and the study was performed in accordance with the Declaration of Helsinki. All participants gave written informed consent, including consent to the research use of their video recordings and to the publication of still images taken from them.

Competing interests

The author(s) has/have no competing interests to declare.

References

Ariel, Mira. 1990. *Accessing noun-phrase antecedents* (Croom Helm Linguistics Series). London: Routledge.

Arık, Engin. 2013. Expressions of space in Turkish Sign Language. In Arık, Engin (ed.), *Current directions in Turkish Sign Language research*, 219–242. Newcastle upon Tyne: Cambridge Scholars Publishing.

Bard, Ellen Gurman & Anderson, Anne & Sotillo, Catherine & Aylett, Matthew & Doherty-Sneddon, Gwyneth & Newlands, Alison. 2000. Controlling the intelligibility of referring expressions in dialogue. *Journal of Memory and Language* 42(1). 1–22. https://doi.org/10.1006/jmla.1999.2667

Bates, Douglas & Mächler, Martin & Bolker, Ben & Walker, Steve. 2015. Fitting linear mixed-effects models using **lme4**. *Journal of Statistical Software* 67(1). https://doi.org/10.18637/jss.v067.i01

Bel, Aurora & Ortells, Marta & Morgan, Gary. 2015. Reference control in the narratives of adult sign language learners. *International Journal of Bilingualism* 19(5). 608–624. https://doi.org/10.1177/1367006914527186

Bell, Alan & Brenier, Jason & Gregory, Michelle & Girand, Cynthia & Jurafsky, Dan. 2009. Predictability effects on durations of content and function words in conversational English. *Journal of Memory and Language* 60(1). 92–111. https://doi.org/10.1016/j.jml.2008.06.003

Bodur, Kübra & Fredouille, Corinne & Rauzy, Stéphane & Meunier, Christine. 2025. Exploring the nuances of reduction in conversational speech: lexicalized and non-lexicalized reductions. *Speech Communication* 173. 103268. https://doi.org/10.1016/j.specom.2025.103268

Boudreault, Patrick & Mayberry, Rachel. 2006. Grammatical processing in American Sign Language: Age of first-language acquisition effects in relation to syntactic structure. *Language and Cognitive Processes* 21(5). 608–635. https://doi.org/10.1080/01690960500139363

Brentari, Diane. 1998. *A prosodic model of sign language phonology*. MIT Press.

Börstell, Carl. 2023. Extracting sign language articulation from videos with MediaPipe. In Alumäe, Tanel & Fishel, Mark (eds.), *Proceedings of the 24th Nordic conference on computational linguistics (NoDaLiDa)*. 169–178. Tórshavn, Faroe Islands: University of Tartu Library. https://aclanthology.org/2023.nodalida-1.18.

Börstell, Carl & Schembri, Adam & Crasborn, Onno. 2024. Sign duration and signing rate in British sign language, Dutch Sign Language and Swedish sign language. *Glossa Psycholinguistics* 3(1). https://doi.org/10.5070/G60111915

Cao, Zhe & Hidalgo, Gines & Simon, Tomas & Wei, Shih-En & Sheikh, Yaser. 2021. OpenPose: Realtime multi-person 2D pose estimation using part affinity fields. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 43(1). 172–186. https://doi.org/10.1109/TPAMI.2019.2929257

Chafe, Wallace. 1976. Givenness, contrastiveness, definiteness, subjects, topics, and point of view. In Li, Charles N. (ed.), *Subject and topic*, 25–55. New York, NY: Academic Press.

Cheng, Qi & Mayberry, Rachel. 2021. When event knowledge overrides word order in sentence comprehension: Learning a first language after childhood. *Developmental Science* 24(5). e13073. https://doi.org/https://doi.org/10.1111/desc.13073. https://onlinelibrary.wiley.com

Cormier, Kearsy & Smith, Sandra & Zwets, Martine. 2013. Framing constructed action in British Sign Language narratives. *Journal of Pragmatics* 55. 119–139. https://doi.org/10.1016/j.pragma.2013.06.002

Czubek, Todd. 2017. *A comprehensive study of referring expressions in ASL*. Boston, MA: Boston University dissertation.

Dikyuva, Hasan & Makaroğlu, Bahtiyar & Arık, Engin. 2017. *Turkish Sign Language grammar*. Ankara: TC Aile ve Sosyal Politikalar Bakanlığı.

Dudis, Paul. 2004. Body partitioning and real-space blends. *Cognitive Linguistics* 15(2). https://doi.org/10.1515/cogl.2004.009

Edwards, Jan & Beckman, Mary & Fletcher, Janet. 1991. The articulatory kinematics of final lengthening. *The Journal of the Acoustical Society of America* 89(1). 369–382. https://doi.org/10.1121/1.400674

Erickson, Frederick. 1992. They know all the lines: Rhythmic organization and contextualization in a conversational listing routine. In Auer, Peter & Di Luzio, Aldo (eds.), *The contextualization of language* (Pragmatics & Beyond New Series 22), 365–397. Amsterdam: John Benjamins.

Eroğlu, Ece. 2025. *Pointing in Turkish Sign Language (TİD)*. Istanbul: Boğaziçi University MA thesis.

Ferrara, Lindsay & Anible, Benjamin & Hodge, Gabrielle & Jantunen, Tommi & Leeson, Lorraine & Mesch, Johanna & Nilsson, Anna-Lena. 2023. A cross-linguistic comparison of reference across five signed languages. *Linguistic Typology* 27(3). 591–627. https://doi.org/10.1515/lingty-2021-0057

Flaherty, Molly & Sato, Asha & Kirby, Simon. 2023. Documenting a reduction in signing space in Nicaraguan Sign Language using depth and motion capture. *Cognitive Science* 47(4). e13277. https://doi.org/10.1111/cogs.13277

Fowler, Carol Ann & Housum, Jonathan. 1987. Talkers' signaling of "new" and "old" words in speech and listeners' perception and use of the distinction. *Journal of Memory and Language* 26(5). 489–504. https://doi.org/10.1016/0749-596X(87)90136-7

Frederiksen, Anne Therese & Mayberry, Rachel. 2016. Who's on first? Investigating the referential hierarchy in simple native ASL narratives. *Lingua* 180. 49–68. https://doi.org/10.1016/j.lingua.2016.03.007

Frederiksen, Anne Therese & Mayberry, Rachel. 2019. Reference tracking in early stages of different modality L2 acquisition: Limited over-explicitness in novice ASL signers' referring expressions. *Second Language Research* 35(2). 253–283. https://doi.org/10.1177/0267658317750220

Gahl, Susanne & Yao, Yao & Johnson, Keith. 2012. Why reduce? Phonological neighborhood density and phonetic reduction in spontaneous speech. *Journal of Memory and Language* 66(4). 789–806. https://doi.org/10.1016/j.jml.2011.11.006

Givón, Talmy. 1983. *Topic continuity in discourse*. John Benjamins Publishing Company. https://benjamins.com/catalog/tsl.3.

Google. 2026. Pose landmark detection guide. Last updated May 28, 2026. https://developers.google.com/edge/mediapipe/solutions/vision/pose_landmarker.

Grice, Herbert Paul. 1975. Logic and conversation. In *Speech acts*, 41–58. Brill.

Gullberg, Marianne. 2006. Handling discourse: Gestures, reference tracking, and communication strategies in early L2. *Language Learning* 56(1). 155–196. https://doi.org/10.1111/j.0023-8333.2006.00344.x

Gundel, Jeanette Kohn & Hedberg, Nancy & Zacharski, Ron. 1993. Cognitive status and the form of referring expressions in discourse. *Language* 69(2). 274–307. https://doi.org/10.2307/416535

Gürer, Aslı & Karabüklü, Serpil & Çavuşoğlu, Kasım Burak. 2025. Duration as a prosodic cue in TİD: Focus realization in the extended domain. *Glossa: a journal of general linguistics* 10(1). https://doi.org/10.16995/glossa.17475

Haspelmath, Martin. 2008. Frequency vs. iconicity in explaining grammatical asymmetries. *Cognitive Linguistics* 19(1). 1–33. https://doi.org/10.1515/COG.2008.001

Hickmann, Maya & Hendriks, Henriëtte & Roland, Françoise & Liang, James. 1996. The marking of new information in children's narratives: a comparison of English, French, German and Mandarin Chinese. *Journal of Child Language* 23(3). 591–619. https://doi.org/10.1017/S0305000900008965

Hoetjes, Marieke & Krahmer, Emiel & Swerts, Marc. 2014. Do repeated references result in sign reduction? *Sign Language & Linguistics* 17(1). 56–81. https://doi.org/10.1075/sll.17.1.03hoe

Janzen, Terry. 2004. Space rotation, perspective shift, and verb morphology in ASL. *Cognitive Linguistics* 15(2). https://doi.org/10.1515/cogl.2004.006

Karadöller, Dilay & Sümer, Beyza & Özyürek, Aslı. 2017. Effects of delayed language exposure on spatial language acquisition by signing children and adults. In Gunzelmann, Glenn & Howes, Andrew & Tenbrink, Thora & Davelaar, Eddy J. (eds.), *Proceedings of the 39th Annual Conference of the Cognitive Science Society*, 2372–2376. Austin, TX: Cognitive Science Society. https://escholarship.org/uc/item/0s0268zt.

Kayabaşı, Demet & Gökgöz, Kadir. 2023. The causative-inchoative alternation and age-of-acquisition effects on multi-predicate constructions in Turkish Sign Language. *Language Learning and Development* 19(2). 125–150. https://doi.org/10.1080/15475441.2022.2047690

Law No. 5378. 2005. Law no. 5378 on persons with disabilities (Engelliler Hakkında Kanun). Official Gazette of the Republic of Türkiye. https://www.resmigazete.gov.tr.

Lenth, Russell & Piaskowski, Julia. 2026. *emmeans: Estimated marginal means, aka least-squares means*. https://doi.org/10.32614/CRAN.package.emmeans. R package version 2.0.3

Levy, Roger. 2008. Expectation-based syntactic comprehension. *Cognition* 106(3). 1126–1177. https://doi.org/10.1016/j.cognition.2007.05.006

Lindblom, B. 1990. *Explaining phonetic variation: A sketch of the h&h theory* 403–439. Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-94-009-2037-8_16. https://doi.org/10.1007/978-94-009-2037-8_16

Lugaresi, Camillo & Tang, Jiuqiang & Nash, Hadon & McClanahan, Chris & Uboweja, Esha & Hays, Michael & Zhang, Fan & Chang, Chuo-Ling & Yong, Ming Guang & Lee,

Juhyun & Chang, Wan-Teh & Hua, Wei & Georg, Manfred & Grundmann, Matthias. 2019. MediaPipe: A framework for building perception pipelines. https://doi.org/10.48550/arXiv.1906.08172

Mauk, Claude & Tyrone, Martha. 2012. Location in ASL: Insights from phonetic variation. *Sign Language & Linguistics* 15(1). 128–146. https://doi.org/10.1075/sll.15.1.06mau

Mauk, Claude Edward. 2003. *Undershoot in two modalities: Evidence from fast speech and fast signing*: The University of Texas at Austin dissertation.

Mayberry, Rachel. 2007. When timing is everything: Age of first-language acquisition effects on second-language learning. *Applied Psycholinguistics* 28(3). 537–549. https://doi.org/10.1017/S0142716407070294

Mitchell, Ross & Karchmer, Michael. 2004. When parents are deaf versus hard of hearing: Patterns of sign use and school placement of deaf and hard-of-hearing children. *The Journal of Deaf Studies and Deaf Education* 9(2). 133–152. https://doi.org/10.1093/deafed/enh017

Nuhbalaoglu, Derya. 2018. *Comprehension and production of referential expressions in German Sign Language and Turkish Sign Language: An empirical approach*. Göttingen: Georg-August-Universität Göttingen dissertation.

Östling, Robert & Börstell, Carl & Courtaux, Servane. ???? Visual iconicity across sign languages: Large-scale automated video analysis of iconic articulators and locations. *Frontiers in Psychology* 9.

Pluymaekers, Mark & Ernestus, Mirjam & Baayen, Rolf Harald. 2005. Lexical frequency and acoustic reduction in spoken Dutch. *The Journal of the Acoustical Society of America* 118(4). 2561–2569. https://doi.org/10.1121/1.2011150

Prince, Ellen. 1978. A comparison of wh-clefts and it-clefts in discourse. *Language* 54(4). 883–906. https://doi.org/10.2307/413238

Puupponen, Anna & Wainio, Tuija & Burger, Birgitta & Jantunen, Tommi. 2015. Head movements in Finnish Sign Language on the basis of motion capture data: A study of the form and function of nods, nodding, head thrusts, and head pulls. *Sign Language & Linguistics* 18(1). 41–89. https://doi.org/10.1075/sll.18.1.02puu

Sandler, Wendy. 1999. Cliticization and prosodic words in a sign language. In Hall, Tracy Alan & Kleinhenz, Ursula (eds.), *Studies on the phonological word* (Current Issues in Linguistic Theory 174), 223–254. Amsterdam: John Benjamins. https://doi.org/10.1075/cilt.174.09san

Selting, Margret. 2007. Lists as embedded structures and the prosody of list construction as an interactional resource. *Journal of Pragmatics* 39(3). 483–526. https://doi.org/10.1016/j.pragma.2006.07.008

Sevgi, Hande & Gökgöz, Kadir. 2023. Classifiers, argument expression, and age of acquisition effects in Turkish Sign Language (TİD). *Sign Language & Linguistics* 26(1). 1–36. https://doi.org/10.1075/sll.21006.sev

Sorace, Antonella. 2011. Pinning down the concept of "interface" in bilingualism. *Linguistic Approaches to Bilingualism* 1(1). 1–33. https://doi.org/10.1075/lab.1.1.01sor

Sperber, Dan & Wilson, Deirdre. 1986. *Relevance: Communication and cognition*, vol. 142. Harvard University Press Cambridge, MA.

Stamp, Rose & Cohn, David & Hel-Or, Hagit & Sandler, Wendy. 2024a. Kinect-ing the dots: Using motion-capture technology to distinguish sign language linguistic from gestural expressions. *Language and Speech* 67(1). 255–276. https://doi.org/10.1177/00238309231169502

Stamp, Rose & Dachkovsky, Svetlana & Hel-Or, Hagit & Cohn, David & Sandler, Wendy. 2024b. A kinematic study of phonetic reduction in a young sign language. *Journal of*

Phonetics 104. 101311. https://doi.org/10.1016/j.wocn.2024.101311
Stokoe, William. 1960. Sign language structure: An outline of the visual communication systems of the American deaf. In *Studies in linguistics: Occasional papers*, Washington, DC: Gallaudet University Press.
Taşçı, Süleyman & Göksel, Aslı. 2018. Native compounds in TİD: A classification based on headedness. *Dilbilim Araştırmaları Dergisi* 29(1). 1–28.
Tily, Harry & Piantadosi, Steven. 2009. Refer efficiently: Use less informative expressions for more predictable meanings. In *Proceedings of the workshop on the production of referring expressions: Bridging the gap between computational and empirical approaches to reference*. 1–8. Amsterdam, The Netherlands.
Toole, Janine. 1996. The effect of genre on referential choice. In Fretheim, Thorstein & Gundel, Jeanette Kohn (eds.), *Reference and referent accessibility* (Pragmatics & Beyond New Series), 263. John Benjamins Publishing Company. https://doi.org/10.1075/pbns.38.16too
Tyrone, Martha & Mauk, Claude. 2010. Sign lowering and phonetic reduction in American Sign Language. *Journal of Phonetics* 38(2). 317–328. https://doi.org/10.1016/j.wocn.2010.02.003
Tyrone, Martha & Mauk, Claude. 2012. Phonetic reduction and variation in American Sign Language: A quantitative study of sign lowering. *Laboratory Phonology* 3(2). https://doi.org/10.1515/lp-2012-0019
Wilbur, Ronnie & Malaia, Evie & Shay, Robin. 2012. Degree modification and intensification in American Sign Language adjectives. In *Logic, language and meaning: 18th Amsterdam colloquium, Amsterdam, the Netherlands, december 19-21, 2011, revised selected papers*. 92–101. Springer.
Yaşar, Aysemin. 2025. *Complexity of telicity marking in Turkish Sign Language (TİD)*. Istanbul: Boğaziçi University MA thesis.
Zeshan, Ulrike. 2003. Aspects of Türk İşaret Dili (Turkish Sign Language). *Sign Language & Linguistics* 6(1). 43–75. https://doi.org/10.1075/sll.6.1.04zes
Zipf, George Kingsley. 1949. *Human behavior and the principle of least effort*. Cambridge, MA: Addison-Wesley.

A Participant outlier analysis

Prior to the main analysis, we assessed whether any participant produced an atypical distribution of RE types that could disproportionately influence the model estimates. One participant (Participant 34), despite having deaf caregivers, had primarily used homesign rather than TİD during childhood. Because this acquisition profile does not meet the DoD inclusion criterion of TİD exposure from birth, data from Participant 34 were excluded from all analyses reported in the main text. Consistent with that profile, this participant showed an unusually high rate of CA use (78% of all REs, compared to a DoD group mean of 24%) and an unusually low rate of nominal use (6%, compared to a DoD group mean where nominals were the dominant introduction strategy), and a Cook's distance diagnostic using a binary nominal vs. non-nominal GLMM corroborated this participant's outsized influence (Cook's distance ≈ 0.6, far exceeding the $4/N \approx 0.14$ cutoff commonly used for influence screening; no other participant was flagged by this diagnostic). As a robustness check, we verified that the exclusion did not change the direction or significance of the key group effects. We note that one retained DoD participant (Participant 13) also lists homesign as the home language; unlike Participant 34, however, this signer was exposed to TİD from birth through the surrounding deaf family and community (AoA 0–3) and used TİD as a primary childhood language, and her RE distribution fell within the DoD group range on the influence diagnostic.

B Interaction style analysis

To assess whether narrating to a camera versus a native addressee impacted the discourse data, we compared Accessibility Scores across interaction styles using a linear mixed-effects model with Group, Discourse context, and Interaction style as fixed effects and random intercepts for Participant and Stimulus. The effect of Interaction [Camera] was not significant ($\beta = -0.02$, $p = 0.514$). Because accessibility scores are computed from discourse structure, this null result speaks only to the comparability of narrative organization across interaction styles; it does not test whether interaction style affected the kinematic measures themselves, a limitation discussed in Section 6. Full model results are shown in Table 5. The significant Group [DoD] coefficient in that table ($\beta = 0.08$, $p = 0.001$) indicates that DoD signers' REs had slightly higher accessibility scores on average; this is a discourse-distribution effect and should not be confused with the non-significant Group main effects on RE-level phonetic measures reported in Section 4.3.

Predictors	Estimates	CI	p
(Intercept)	1.49	1.37 – 1.62	<0.001
Group [DoD]	0.08	0.03 – 0.13	0.001
Discourse [Introduction]	-2.28	-2.31 – -2.24	<0.001
Discourse [Maintenance]	2.68	2.65 – 2.71	<0.001
Interaction [Camera]	-0.02	-0.09 – 0.05	0.514

Table 5: Discourse Model results for Accessibility Score (Interaction Style)

C Accessibility score worked examples

The following examples illustrate how the accessibility score was calculated for each annotated referring expression, regardless of RE type. Accessibility scoring was applied to the full discourse annotation (nominals, classifiers, CA, verbs, etc.); only the subsequent kinematic analyses restrict attention to the overt nominals CAT and MOUSE. All three examples belong to the same narrative and represent three consecutive clauses from a signer's first narrative. Each example contains tiers for gloss, the referent's syntactic role, its discourse context and type, scores for three primary factors (distance and unity, topicality, and competition), and the total accessibility score.

(12) Clause 1





CAT MOUSE CHASE

Referent's Syntactic Role:	Subject	Object
Discourse Context:	Introduction	Introduction
RE Type:	Nominal	Nominal
Distance and Unity Score:	0	0
Topicality Score:	0	0
Competition Score:	0	0
Total Accessibility Score:	0	0

Translation: 'Tom is chasing Jerry.'

In (12), the signer introduces Tom and Jerry with the nominals CAT and MOUSE. Because they lack prior mentions in the recent discourse context, both referents receive a distance/unity score of 0 and a topicality score of 0. As there are no competing referents requiring attention, their competition score is also 0, yielding a total accessibility score of 0 for both referents.

(13) Clause 2




TUNNEL JUMP

Referent's Syntactic Role:	Adjunct	Subject
Discourse Context:	Introduction	Maintenance
RE Type:	Classifier	Classifier (Right Hand)
Distance and Unity Score:	0	3
Topicality Score:	0	1
Competition Score:	-2	-1
Total Accessibility Score:	-2	3

Translation: 'Jerry is jumping into a tunnel.'

The example in (13) features a new referent (TUNNEL) introduced via a static classifier alongside the maintained subject Jerry, realized here not with an overt nominal but with a whole-entity classifier predicate (JUMP). TUNNEL receives an initial distance/unity score

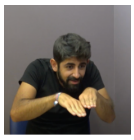
of 0, and owing to two preceding competing referents (Tom and Jerry), its competition score drops to −2, resulting in a total accessibility score of −2. In contrast, the maintained referent Jerry has been mentioned in the immediately preceding clause, securing a distance/unity score of 3 and a topicality increase of 1. With only one intervening competitor (TUNNEL), Jerry's final accessibility score is 3. This clause is included precisely to show that the scoring algorithm applies to non-nominal REs as well; such tokens enter the discourse-distribution analyses but not the kinematic models.

Finally, (14) illustrates the nominal re-introduction of both CAT and MOUSE:

(14) Clause 3


CAT


MOUSE


CRAWL.CA

Referent's Syntactic Role:	Subject	Subject
Discourse Context:	Re-introduction	Re-introduction
RE Type:	Nominal	Nominal (Right Hand)
Distance and Unity Score:	2	2
Topicality Score:	1	1
Competition Score:	-2	-1
Total Accessibility Score:	1	2

Translation: 'Tom and Jerry start crawling (in the tunnel).'

Tom's last mention occurred in Clause 1 (in the same narrative context but not in the same or immediately preceding clause), so criterion (b) assigns a distance/unity score of 2. Jerry's last mention (the classifier predicate JUMP) does occur in the immediately preceding clause; however, the mention of the competing referent CAT earlier in Clause 3 intervenes between Jerry's previous and current mention. This breaks the subject continuity required for maintenance coding (Section 3.3), so Jerry is coded as re-introduced; because the interruption also voids the unbroken adjacency presupposed by criterion (a) (see Section 3.4.1), distance/unity is likewise scored by criterion (b) (score 2). Both referents receive a topicality score of 1. The CAT referent is penalized −2 for facing two intervening competitors between its mentions (TUNNEL and Jerry), yielding a final score of 1, whereas MOUSE faces only one competitor (CAT), yielding an accessibility score of 2.

D Polynomial accessibility models

To formally characterize the shape of the accessibility-phonetics relationship, we fitted polynomial mixed-effects models extending the linear models in Table 3. The random-effects structure matched those models, as did the fixed-effect skeleton for each measure (including RE type only where the linear models retained it: duration and signing space size, but not hand distance). Table 6 reports only the polynomial terms, as the remaining coefficients did not differ substantively from those in the linear model.

Predictors	Duration (log) Estimate [CI]	Hand Distance (log) Estimate [CI]	Signing Space Size (log) Estimate [CI]
Accessibility^2 (quadratic)	n.s.	0.17 [0.01, 0.32]*	0.17 [0.03, 0.30]*
Accessibility^3 (cubic)	0.21 [0.06, 0.36]**	n.s.	n.s.

Note. * $p < 0.05$, ** $p < 0.01$. n.s. = not significant (term included in model but $p > .05$).

Table 6: Polynomial (quadratic and cubic) accessibility terms from mixed-effects models for each phonetic measure. All other terms as in Table 3.

The significant cubic term for duration indicates two turning points along the accessibility scale, consistent with finer temporal sensitivity spanning introduction-, re-introduction-, and maintenance-associated regions of that continuum (rather than with pairwise-distinct categorical Discourse levels). The significant quadratic (but not cubic) terms for hand distance and signing space size indicate a single turning point, consistent with these measures tracking a binary new-versus-predictable distinction with a plateau at higher accessibility levels.